

The Epoch of Reionization: Observational State of the Art

and a Catalogue of Ionizing and Heating Agents

Literature synthesis prepared for L. Carvalho

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Abstract

This document summarises the current observational picture of the Epoch of Reionization (EoR) and catalogues the candidate agents of hydrogen/helium ionization and of intergalactic medium (IGM) heating, together with the quantitative criteria the literature actually uses to decide whether a given source population is *dominant*, *significant* or *negligible*. It is built from a corpus of 60 refereed/preprint papers selected for citation impact and recency (1999–2026), stored as PDFs in the companion folders **Constraints/** (25 papers) and **Agents/** (35 papers). Every numerical value quoted below is traceable to a specific paper in that corpus; nothing has been invented, and quantities that could not be sourced are explicitly flagged. Section 6 lists the additional topics I encountered but did *not* include, pending your decision.

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1 Scope, corpus and method

1.1 How the corpus was assembled

Candidate papers were surfaced by (i) programmatic queries to the arXiv API (`astro-ph.C0`) across the source-physics and constraint-physics sub-topics named in the task specification, and (ii) targeted retrieval of landmark papers by arXiv identifier. Every identifier was verified against the arXiv API before download (title, author list and posting date returned by the API are stored

in `corpus_metadata.json`). All 60 PDFs downloaded successfully; no entry in the bibliography is a paper I did not retrieve.

Papers were partitioned into the two requested folders by their *primary* scientific claim:

- **Constraints/** — papers whose main result is a measurement of, or an inference on, a parameter that characterises the EoR (τ , $\langle x_{\text{HI}} \rangle(z)$, $\langle \Gamma_{\text{HI}} \rangle$, λ_{mfp} , T_0 , T_{S} , bubble size, z_{end} , z_{50}).
- **Agents/** — papers whose main result is the identification or quantification of the *contribution* of a source population or a physical channel (LyC photons, X-rays, cosmic rays, Ly α photons, secondary electrons) to ionization and/or heating.

Several papers legitimately straddle the boundary (e.g. HERA Collaboration et al. [1] simultaneously reports a power-spectrum upper limit and an L_{X} /SFR constraint). In those cases the folder assignment follows the paper’s *stated* headline result; the full corpus table (Appendix A) records the assignment explicitly.

1.2 Provenance conventions used throughout

- Uncertainties are reproduced exactly as quoted by the source paper, together with the confidence level the source states (68%, 95%, 99%, or “1 σ ”). Where a source gives no confidence level I say so.
- Citation counts are *not* used as evidence for any physical claim. Where I mention them (Appendix A) they come from the Semantic Scholar API, retrieved 8 September 2026, and are known to undercount astronomy literature relative to NASA ADS; treat them as a relative ranking only.
- BibTeX keys are built from the *arXiv posting year*, which for some entries differs from the journal year (e.g. Atek2023 is Nature 2024; HERA2022 is ApJ 2023).
- **[flagged]** marks a statement I could not verify from the corpus.
- **Figures.** Every figure in this document is the *original published figure*, extracted directly from the PDF in **Agents/** or **Constraints/** at 300 dpi with `pdftocairo`; none has been redrawn, recoloured or replotted. Each caption names the source paper and figure number. They are reproduced here for private research use; any onward use requires the publisher’s permission. The extracted PNGs live in **figures/**.

1.3 Source-consistency audit

Every numerical value in this document has been checked against the text of the PDF it cites, using the script `verify_claims.py` (shipped alongside). The procedure: 116 numeric claims were encoded as (citation key, required numeric tokens); each paper PDF was converted to text with `pdftotext`; tokens were matched after Unicode/whitespace normalisation; and the surrounding ± 95 characters of every match were read to confirm the number is used in the sense claimed here, not merely present somewhere in the paper.

Result. 114 of 116 claims verified on the first pass. The audit found **two substantive numerical errors**, both traceable to a single root cause, and both now corrected:

1. **Chaubal et al. [2] (SPT-3G D1).** The values originally quoted ($D_{3000}^{\text{kSZ}} = 1.75 \pm 0.86$, $D_{3000}^{\text{tSZ}} = 4.91 \pm 0.37 \mu\text{K}^2$, $\Delta^{50} z_{\text{re}} < 3.8$, $\Delta^{90} z_{\text{re}} < 6.1$) are the *v1* values. The PDF in **Constraints/** is v2, which reports 1.98 ± 0.87 , $4.98 \pm 0.35 \mu\text{K}^2$, $\Delta^{50} z_{\text{re}} < 4.2$ and $\Delta^{90} z_{\text{re}} < 7.0$. Section 2.8 now quotes v2.
2. **Rosdahl et al. [3] (SPHINX).** The binary-star boost to f_{esc} was quoted as $3.5\times$; the v2 PDF states “about 3 times higher”, with a luminosity-weighted 8.5% (binaries) versus 2.7% (singles), i.e. a factor ≈ 3.1 . Section 3 now quotes v2.

Root cause. The arXiv API returns, for some entries, the abstract of an *earlier* version than the identifier it reports and than the PDF it serves. Both errors entered because a value was taken from an API abstract rather than from the downloaded paper. A systematic sweep comparing every API abstract against its PDF flagged exactly two further cases: Chaubal et al. [2] (the error above) and Simmonds et al. [4], whose listing abstract says 15 721 galaxies where the v2 PDF says 14 652 — a number this document never quoted, so no correction was needed. **Two provenance improvements** were also made: $h = 0.6736$ is now stated as derived from the published $H_0 = (67.36 \pm 0.54) \text{ km s}^{-1} \text{ Mpc}^{-1}$ (the bare value h does not appear in Planck Collaboration et al. 5), and the Naidu et al. [6] “oligarch” definition now carries its $\log(M_\star/M_\odot) > 8$ condition, which had been dropped.

Limitation of the audit. Token matching cannot detect a number that is correct in the paper but attached to the wrong claim here *if* the same digits occur elsewhere in that paper. The Rosdahl et al. [3] error was of exactly this type and was caught only by reading contexts by hand. I read the matched context for all 116 claims, but this is a manual step and I cannot certify it exhaustive.

1.4 Independent numerical cross-checks performed

Two checks were run with `sympy`/Python (script: `crosscheck_reionization.py`, shipped alongside this file; all inputs cited, no free parameters):

1. **Dimensional consistency** of the reionization ODE (Eq. 3) and of the recombination time (Eq. 5): both pass under `sympy.physics.units.systems.si.dimsys_SI`.
2. **Order-of-magnitude closure of the photon budget.** Using $\Omega_b h^2 = 0.02237$, $Y_p = 0.2454$ and $H_0 = (67.36 \pm 0.54) \text{ km s}^{-1} \text{ Mpc}^{-1}$, i.e. $h = 0.6736$ [5, TT,TE,EE+lowE+lensing], the comoving mean hydrogen density is $\langle n_H \rangle = 1.895 \times 10^{-7} \text{ cm}^{-3} = 5.566 \times 10^{66} \text{ cMpc}^{-3}$. The measured ionizing emissivity $\dot{n}_{\text{ion}} \simeq 0.6 \times 10^{51} \text{ s}^{-1} \text{ cMpc}^{-3}$ at $4.9 \leq z \leq 6.0$ [7] therefore corresponds to 3.4 ionizing photons per hydrogen atom per Gyr. Independently, Eq. 4 with the fiducial $f_{\text{esc}} = 0.2$, $\log_{10} \xi_{\text{ion}} = 53.14 [\text{s}^{-1}/(\text{M}_\odot \text{ yr}^{-1})]$ of Robertson et al. [8] and $\rho_{\text{SFR}} \simeq 0.01\text{--}0.02 \text{ M}_\odot \text{ yr}^{-1} \text{ cMpc}^{-3}$ gives $\dot{n}_{\text{ion}} = 2.8\text{--}5.5 \times 10^{50} \text{ s}^{-1} \text{ cMpc}^{-3}$. The two routes agree to within a factor $\lesssim 2$, which is the expected level given the f_{esc} and ρ_{SFR} uncertainties.

Not cross-checked: every value quoted directly from a paper’s abstract, table or text (Sections 2–3) is reproduced as published and has *not* been re-derived here. Where two papers in the corpus disagree, the disagreement is reported rather than resolved (Section 5).

2 What observations establish about the EoR

2.1 The cosmological framework and the integral constraint

Thomson scattering of CMB photons off free electrons produced by reionization imprints an optical depth

$$\tau = c \sigma_T \int_0^{z_{\text{max}}} \frac{n_e(z)}{(1+z)H(z)} dz, \quad (1)$$

which is an *integral* of the ionization history and therefore constrains its total but not its shape. The current values are

- $\tau = 0.054 \pm 0.007$ (68% CL), from the final full-mission *Planck* TT,TE,EE+lowE+lensing analysis [5].
- $\tau = 0.0566^{+0.0053}_{-0.0062}$ (68% CL) from large-scale polarization alone, and $\tau = 0.059 \pm 0.006$ (68% CL) in combination, using the improved SRoll2 HFI map-making [9]. This corresponds to a mid-point reionization redshift $z_{\text{re}} = 8.14 \pm 0.61$ (68% CL) *within a parametrised (tanh) reionization model* — the model dependence of z_{re} is essential and is frequently under-stated.

- An independent, astrophysics-anchored route gives $\tau_{\text{reio}} = 0.067 \pm 0.011$: Montero-Camacho and Mao [10] combine CMB *excluding* large-scale polarization, BAO, quasar damping-wing $\langle x_{\text{HI}} \rangle$ measurements and dark-pixel limits within a Gompertzian reionization parametrisation, and find agreement with the polarization-based value.

2.2 The end of reionization: $z_{\text{end}} \simeq 5.2\text{--}5.3$

Four independent Ly α -forest diagnostics now converge on a *late* end to reionization.

Dark-pixel fraction (nearly model independent). Any pixel with detected transmission must be highly ionized; dark pixels may be either. The dark fraction is therefore an upper limit on $\langle x_{\text{HI}} \rangle$ requiring no assumption about the ionizing background or the quasar continuum [11]:

- $\langle x_{\text{HI}} \rangle \leq 0.06 + 0.05$ (1σ) at $z = 5.9$ and $\langle x_{\text{HI}} \rangle \leq 0.04 + 0.05$ (1σ) at $z = 5.6$, from 22 quasars [11];
- updated with 34 deep XQR-30 spectra using Ly β +Ly γ [12]: $\langle x_{\text{HI}} \rangle \leq \{0.030 + 0.048, 0.095 + 0.037, 0.191 + 0.056, 0.199 + 0.087\}$ (1σ) at $\bar{z} = \{5.481, 5.654, 5.831, 6.043\}$. The authors conclude that the bulk of reionization is finished by $z > 6$, while a “soft landing” extending to $z \sim 5.4$ remains allowed.

Opacity fluctuations. Becker et al. [13] discovered a $\sim 110 h^{-1}$ Mpc trough with $\tau_{\text{eff}} \gtrsim 7$ extending to $z \simeq 5.5$ towards ULAS J0148+0600, and showed that the sightline-to-sightline scatter at $z \simeq 5.6\text{--}5.8$ requires volume-weighted $\langle x_{\text{HI}} \rangle$ fluctuations a factor ≥ 3 larger than density fluctuations alone can produce. By $z \sim 5$ the scatter is fully explained by the density field.

Ly α transmission statistics. With 67 sightlines at $z > 5.5$ (largely XQR-30), Bosman et al. [14] find homogeneous-UVB simulations consistent with the data at $z \leq 5.2$ ($< 1\sigma$), in mild tension at $z = 5.3$ ($\sim 2.5\sigma$), and *ruled out* at $z \geq 5.4$ ($> 3.5\sigma$). Reionization-related fluctuations therefore persist to at least $z = 5.3$ ($t = 1.1$ Gyr).

Dark gaps. Using 55 quasars, Zhu et al. [15] measure the fraction of sightlines with gaps longer than $30 h^{-1}$ Mpc: $F_{30} \simeq 0.9, 0.6, 0.15$ at $z = 6.0, 5.8, 5.6$, with nine ultra-long ($L > 80 h^{-1}$ Mpc) gaps at $z < 6$ and long gaps persisting to $z \simeq 5.3$.

Radiative-transfer modelling of the emissivity, mean free path and neutral fraction jointly favours completion at $z \simeq 5.2$ [7].

2.3 The mid-point and the neutral fraction at $z = 6\text{--}10$

Table 1: Volume-averaged neutral hydrogen fraction $\langle x_{\text{HI}} \rangle$ from the corpus. Uncertainties and confidence intervals are exactly as quoted by the source.

Redshift	$\langle x_{\text{HI}} \rangle$	Probe	Reference
5.481	$\leq 0.030 + 0.048$ (1σ)	Ly β /Ly γ dark pixels	Davies et al. [12]
5.654	$\leq 0.095 + 0.037$ (1σ)	Ly β /Ly γ dark pixels	Davies et al. [12]
5.6	$\leq 0.04 + 0.05$ (1σ)	Ly α /Ly β dark pixels	McGreer et al. [11]
5.831	$\leq 0.191 + 0.056$ (1σ)	Ly β /Ly γ dark pixels	Davies et al. [12]
5.9	$\leq 0.06 + 0.05$ (1σ)	Ly α /Ly β dark pixels	McGreer et al. [11]
6.00	$0.174^{+0.093}_{-0.109}$	Forward-modelled τ_{eff}	Gaikwad et al. [7]
6.043	$\leq 0.199 + 0.087$ (1σ)	Ly β /Ly γ dark pixels	Davies et al. [12]
6.10	$0.21^{+0.17}_{-0.07}$ (ATON)	Quasar damping wing (stacked)	Ďurovčíková et al. [16]

Redshift	$\langle x_{\text{HI}} \rangle$	Probe	Reference
6.46	$0.21^{+0.33}_{-0.07}$ (ATON)	Quasar damping wing (stacked)	Žurovčíková et al. [16]
6.87	$0.37^{+0.17}_{-0.17}$ (ATON)	Quasar damping wing (stacked)	Žurovčíková et al. [16]
~ 7	$0.59^{+0.11}_{-0.15}$	Ly α EW of LBGs (Bayesian)	Mason et al. [17]
7.09	0.48 ± 0.26 (68% cred.)	Damping wing, ULAS J1120+0641	Davies et al. [18]
$7.12^{+0.06}_{-0.08}$	$0.53^{+0.18}_{-0.47}$	JWST galaxy damping wings	Umeda et al. [19]
7.54	$0.60^{+0.20}_{-0.23}$ (68% cred.)	Damping wing, ULAS J1342+0928	Davies et al. [18]
$9.91^{+1.49}_{-1.15}$	$0.92^{+0.08}_{-0.10}$	JWST galaxy damping wings	Umeda et al. [19]

Global inferences that combine these probes give:

- EoR mid-point in $6.9 < z < 7.6$, with large spatial fluctuations in the IGM ionization state [20].
- $z_{50} = 7.16^{+0.15}_{-0.12}$ and a duration $\Delta z_{\text{re}} < 1.8$ (95% credibility) from a joint Bayesian analysis of *Planck*+SPT CMB power spectra, Lyman-line $\langle x_{\text{HI}} \rangle$ data, and 21-cm upper limits from HERA, LOFAR and the MWA [21]. This is model dependent, as the authors stress.
- An observationally-derived ionizing budget from 1428 JWST/NIRSpec spectra yields an $\langle x_{\text{HI}} \rangle(z)$ history ending at $z \sim 5.8$ [22].

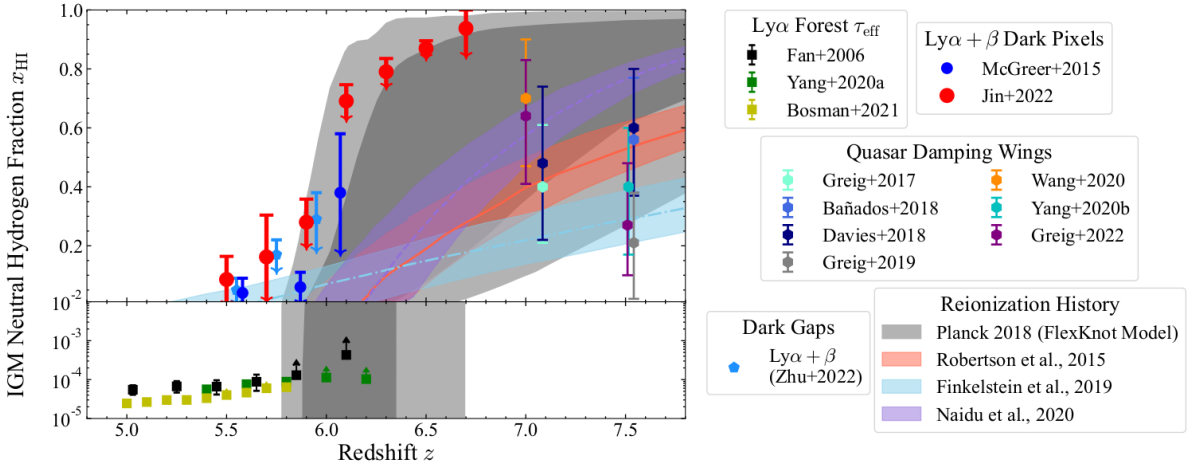


Figure 1: **The reionization history as constrained by quasar absorption.** All data points are quasar-absorption constraints; the dark (light) grey band is the 1σ (2σ) CMB constraint from Planck Collaboration et al. [5]; the red, blue and purple bands are the 1σ histories of the Robertson et al. [8], Finkelstein et al. [23] and Naidu et al. [6] models respectively. Note the two-panel split: below $z \simeq 6$ the forest measures $x_{\text{HI}} \sim 10^{-5}$ – 10^{-4} (lower panel, log scale), so the transition spans four orders of magnitude in neutral fraction over $\Delta z \lesssim 1$. The model bands diverge most where the data are weakest, $z \gtrsim 7$.

Reproduced from Fan et al. [20], Figure 16 (figure therein adapted from Jin et al. 2022).

Caveat on the damping-wing systematic. Žurovčíková et al. [16] report $\langle x_{\text{HI}} \rangle$ values that differ substantially depending on which simulation grid is used to interpret the same stacked spectra: at $\langle z \rangle = 6.46$ they obtain $0.21^{+0.33}_{-0.07}$ with ATON but $0.57^{+0.26}_{-0.47}$ with CROC. The simulation suite is thus a leading systematic, not a detail. They simultaneously constrain the mean quasar lifetime to $t_Q \lesssim 7$ Myr in all bins.

2.4 The ionizing background: $\langle\Gamma_{\text{HI}}\rangle$, λ_{mfp} and $\dot{n}_{\text{ion}}$

The three quantities are linked: in the local-source approximation the photoionization rate scales as $\Gamma_{\text{HI}} \propto \epsilon_{\nu} \lambda_{\text{mfp}}$, so a measurement of two determines the third. Formally,

$$\Gamma_{\text{HI}} = \int_{\nu_{912}}^{\infty} \frac{4\pi J_{\nu}}{h\nu} \sigma_{\text{HI}}(\nu) d\nu. \quad (2)$$

[The proportionality prefactor relating Γ_{HI} to $\epsilon_{\nu} \lambda_{\text{mfp}}$ depends on the assumed spectral index and on the column density distribution; I have not re-derived it, and none of the corpus papers states it in closed form.]

Table 2: Measured properties of the ionizing background from Gaikwad et al. [7], Table 3 (XShooter+ESI quasar sample, forward-modelled τ_{eff} distribution). $\Gamma_{12} \equiv \langle\Gamma_{\text{HI}}\rangle/10^{-12} \text{ s}^{-1}$; λ_{mfp} in $h^{-1} \text{ cMpc}$; $\dot{n}_{\text{ion}}$ in $10^{51} \text{ s}^{-1} \text{ cMpc}^{-3}$. Uncertainties are total 1σ (modelling + cosmic variance + observational). Compare the independent direct measurements of Becker et al. [24]: $\lambda_{\text{mfp}} = 9.09^{+1.62}_{-1.28} \text{ pMpc}$ at $z = 5.1$ and $0.75^{+0.65}_{-0.45} \text{ pMpc}$ at $z = 6.0$ (68% confidence).

z	Γ_{12}	λ_{mfp}	$\langle f_{\text{HI}} \rangle$	$\dot{n}_{\text{ion}}$
4.90	$0.501^{+0.275}_{-0.232}$	$50.1^{+33.1}_{-19.9}$	$2.53^{+0.59}_{-0.44} \times 10^{-5}$	$0.600^{+0.304}_{-0.248}$
5.00	$0.557^{+0.376}_{-0.218}$	$57.4^{+38.1}_{-21.1}$	$2.27^{+0.84}_{-0.45} \times 10^{-5}$	$0.563^{+0.352}_{-0.207}$
5.20	$0.502^{+0.292}_{-0.193}$	$40.8^{+22.3}_{-15.7}$	$2.76^{+0.81}_{-0.56} \times 10^{-5}$	$0.668^{+0.461}_{-0.238}$
5.40	$0.372^{+0.217}_{-0.126}$	$29.2^{+15.4}_{-10.2}$	$3.53^{+15.08}_{-2.46} \times 10^{-3}$	$0.648^{+0.462}_{-0.225}$
5.60	$0.319^{+0.194}_{-0.120}$	$23.0^{+11.7}_{-7.8}$	$1.63^{+2.54}_{-0.83} \times 10^{-2}$	$0.666^{+0.402}_{-0.221}$
5.80	$0.178^{+0.194}_{-0.078}$	$13.2^{+9.2}_{-5.8}$	$9.36^{+6.18}_{-6.39} \times 10^{-2}$	$0.609^{+0.420}_{-0.141}$
6.00	$0.145^{+0.157}_{-0.087}$	$8.32^{+7.53}_{-4.05}$	$1.74^{+0.93}_{-1.09} \times 10^{-1}$	$0.701^{+0.357}_{-0.191}$

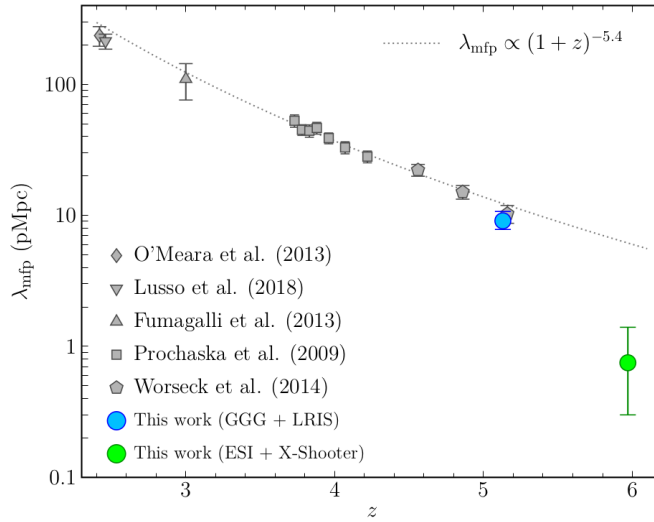


Figure 2: **The collapse of the mean free path at $z = 6$.** Direct λ_{mfp} measurements against the $\lambda_{\text{mfp}} \propto (1+z)^{-5.4}$ power law fitted over $2.44 < z < 5.16$ by Worseck et al. (2014). The $z = 5.1$ point (blue) sits on the extrapolation; the $z = 6.0$ point (green) falls a factor ~ 7 below it. This is the single measurement that most strongly forces reionization to end late.

Reproduced from Becker et al. [24], Figure 8.

Three physical statements follow directly from Table 2, which Figure 3 plots:

1. Between $z \sim 5$ and $z \sim 6$, λ_{mfp} falls by a factor ~ 6 and $\langle f_{\text{HI}} \rangle$ rises by a factor $\sim 10^4$ [7]. Becker et al. [24] independently show the $z = 6.0$ mean free path lies well below

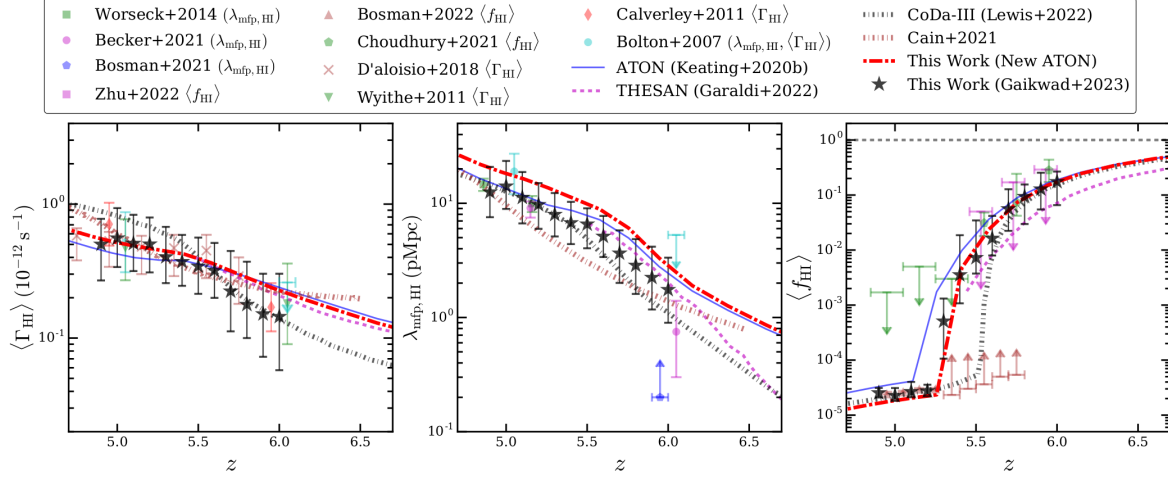


Figure 3: **The three coupled quantities of Table 2.** Left to right: the photoionization rate $\langle \Gamma_{\text{HI}} \rangle$, the mean free path λ_{mfp} , and the neutral fraction $\langle f_{\text{HI}} \rangle$ over $4.9 \leq z \leq 6.0$ (black stars), against literature measurements and the ATON, THESAN, CoDa-III and Cain et al. simulations. The near-vertical rise of $\langle f_{\text{HI}} \rangle$ between $z = 5.2$ and $z = 6.0$ — four orders of magnitude — is the end of reionization; $\langle \Gamma_{\text{HI}} \rangle$ falls by only a factor ~ 4 across the same interval, because the drop is driven by λ_{mfp} , not by the sources.

Reproduced from Gaikwad et al. [7], Figure 10.

extrapolations from lower redshift (Figure 2), and that neglecting the quasar proximity effect biases it high by a factor ≥ 2 .

2. The emissivity $\dot{n}_{\text{ion}}$ is, by contrast, nearly constant at $\simeq 0.6 \times 10^{51} \text{ s}^{-1} \text{ cMpc}^{-3}$ across $4.9 \leq z \leq 6.0$. The drop in $\langle \Gamma_{\text{HI}} \rangle$ is therefore driven by the collapsing mean free path, not by a collapsing source population.
3. That emissivity corresponds to 3.4 ionizing photons per hydrogen atom per Gyr (my cross-check, Section 1.4) — i.e. the IGM at the end of the EoR is *photon starved*: only a few photons per atom per Hubble time are available, so recombinations (Eq. 5) matter at leading order.

2.5 The thermal state of the ionized IGM

The temperature at mean density, T_0 , and the slope γ of the temperature–density relation $T = T_0 \Delta^{\gamma-1}$ are measured from the Ly α forest. Note carefully: *these measurements constrain the already-ionized gas*. The temperature of the residual *neutral* IGM is constrained only by 21-cm experiments (Section 2.6).

- From the width distribution of transmission spikes in five high-resolution $6 < z \lesssim 6.5$ quasar spectra compared to the Sherwood and Sherwood-Relics suites, Gaikwad et al. [25] obtain $T_0 = (11000 \pm 1600, 10500 \pm 2100, 12000 \pm 2200) \text{ K}$ at $z = (5.4, 5.6, 5.8)$, with γ mildly steeper than isothermal. Spectral resolution $\leq 8 \text{ km s}^{-1}$ is required to resolve the spikes.
- A 400-simulation Cholla suite fit jointly to the H I 1D flux power spectrum in 14 bins over $2.2 \leq z \leq 5.0$ and to the He II Ly α opacity in 5 bins over $2.4 < z < 2.9$ yields the two-peak thermal history of Figure 4 [26]: hydrogen reionization completing by $z \sim 6.0$ with a first peak $T_0 \sim 1.3 \times 10^4 \text{ K}$, followed by He II reionization completing by $z \sim 3.0$ with a second peak $T_0 \sim 1.4 \times 10^4 \text{ K}$.
- Radiative-transfer simulations reach volume-averaged $\langle T \rangle \sim 20,000 \text{ K}$ at $z = 6$ once the volume is essentially fully ionized ($\langle x_{\text{HII}} \rangle = 0.99998$) [27].

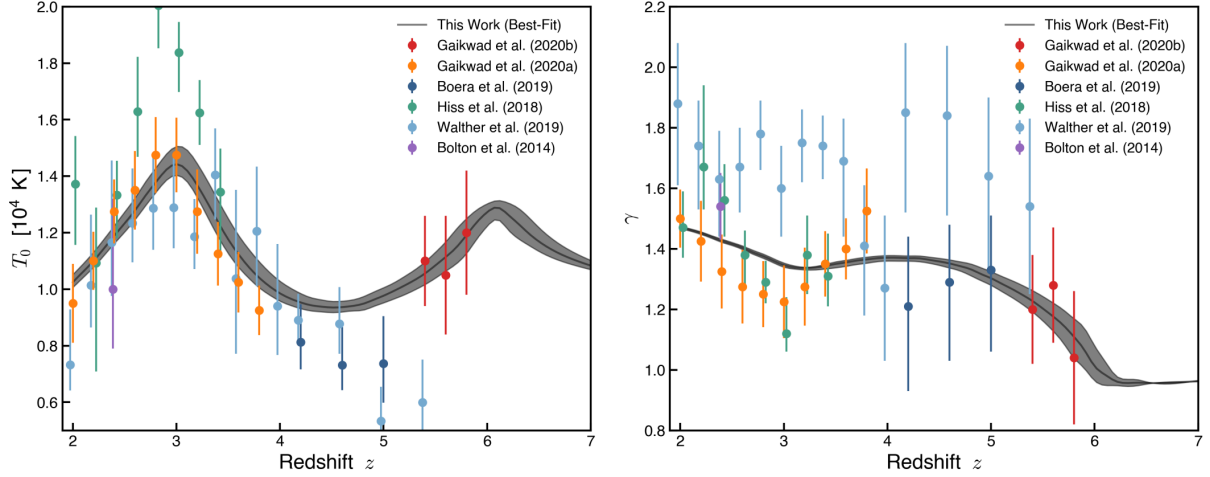


Figure 4: **The two-peak thermal history of the *ionized* IGM.** Redshift evolution of T_0 (left) and the temperature–density slope γ (right) from the best-fit model (black line, 95% band) against Ly α forest measurements. The peak at $z \simeq 6$ is hydrogen reionization; the peak at $z \simeq 3$ is He II reionization by AGN. These are the two heating events that stellar and AGN sources respectively imprint on the IGM — and note that nothing here constrains the *neutral* gas, which is the domain of Figure 5.

Reproduced from Villaseñor et al. [26], Figure 11.

2.6 21-cm constraints on the neutral, pre-reionization IGM

The 21-cm differential brightness temperature depends on the spin temperature T_S through $\delta T_b \propto (1 - T_{\text{CMB}}/T_S)$, so upper limits on the power spectrum translate into *lower* limits on T_S and hence on the heating history. No detection exists; all results below are exclusions.

HERA Phase I (94 nights) [1].

- $\Delta^2(k = 0.34 h \text{ Mpc}^{-1}) \leq 457 \text{ mK}^2$ at $z = 7.9$ and $\Delta^2(k = 0.36 h \text{ Mpc}^{-1}) \leq 3496 \text{ mK}^2$ at $z = 10.4$ (95% confidence).
- Joint two-band fit: $T_S > 11.0 (35.2) \text{ K}$ at $z = 7.9$ and $T_S > 6.2 (26.4) \text{ K}$ at $z = 10.4$ at 95% (68%) credibility. Since $T_{\text{CMB}} = 24.3 \text{ K}$ at $z = 7.9$, “cold reionization” scenarios with enhanced power at $z = 7.9$ are ruled out at $> 95\%$ credibility.
- Corresponding intervals on the kinetic temperature: $3.2 \text{ K} < T_K < 313.2 \text{ K}$ at $z = 10.4$ and $13.0 \text{ K} < T_K < 4768 \text{ K}$ at $z = 7.9$.
- **Heating-source constraint:** $L_{X < 2 \text{ keV}}/\text{SFR} \sim 10^{39.9-41.6} \text{ erg s}^{-1} (\text{M}_\odot \text{ yr}^{-1})^{-1}$ (68% HPD), tightening to $10^{40.4-41.8}$ at 95%, under the prior $L_{X < 2 \text{ keV}}/\text{SFR} < 10^{42}$ (above which X-rays reionize the Universe too quickly). More than 99% of the posterior volume excludes the *local*, metal-enriched L_X –SFR relation continuing to high redshift; the data are consistent with low-metallicity HMXB models. This is the single most direct observational statement in the corpus about *which* agent heated the IGM.

LOFAR (10 nights) [28]. Interpreting the LOFAR upper limits with the GRIZZLY code, models disfavoured at $z = 9.1$ correspond to an averaged ionized-plus-heated fraction below 0.46 ($\lesssim 0.05$ at 68%), an average gas temperature below 44 K (4 K at 68%), and a characteristic heated-region size $\lesssim 14 h^{-1} \text{ Mpc}$ ($\lesssim 3 h^{-1} \text{ Mpc}$ at 68%), at 95% (68%) credibility. The authors emphasise these are *extreme* models driven by rare large ionized/heated regions, and that the 68% interval prefers an excess radio background $> 100\%$ of the CMB at 1.42 GHz while the 95% interval spans the whole prior — i.e. the excess-radio-background degeneracy is not broken.

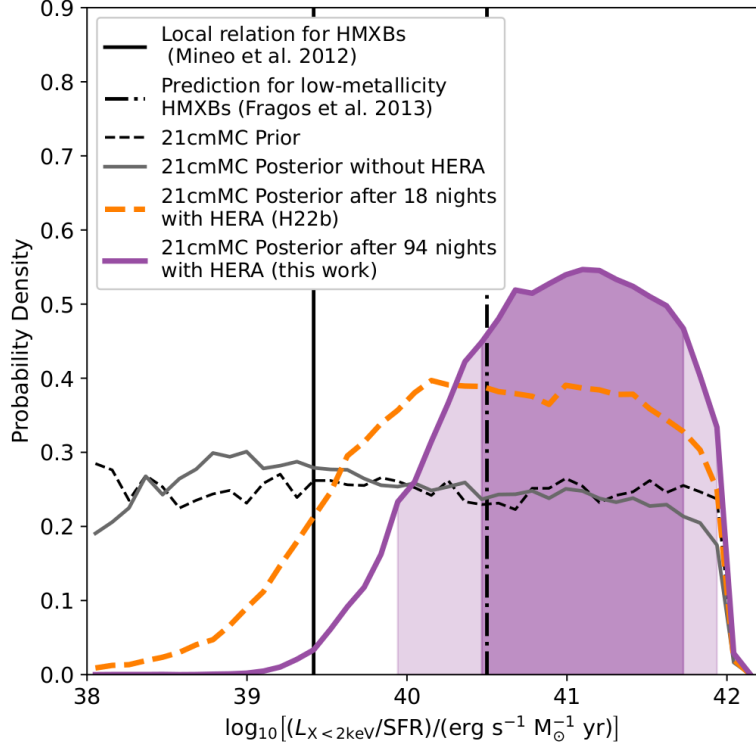


Figure 5: **The strongest direct observational statement about which agent heated the IGM.** Marginalised posterior on the soft X-ray luminosity per unit star formation, $L_{X<2\text{keV}}/\text{SFR}$, with 68% and 95% credible intervals shaded. The solid vertical line is the *local*, metal-enriched HMXB relation (Mineo et al. 2012); the dash-dotted line is the prediction for low-metallicity HMXBs (Fragos et al. 2013). More than 99% of the posterior volume lies above the local relation, so the first galaxies cannot have followed it — the data require low-metallicity X-ray binaries.

Reproduced from HERA Collaboration et al. [1], Figure 28.

2.7 The 21-cm forest: a direct probe of the *neutral* IGM temperature

Absorption by intervening neutral hydrogen in the spectrum of a high- z radio-loud source produces a “21-cm forest”. Because the optical depth scales roughly as $\tau_{21} \propto x_{\text{HI}}/T_{\text{S}}$, the forest constrains the temperature of exactly the phase that the Ly α forest cannot reach (Section 2.5).

- Shao et al. [29] show that the *one-dimensional power spectrum* of the 21-cm forest breaks the otherwise severe degeneracy between warm dark matter (which suppresses small-scale structure) and early heating (which suppresses the absorption directly), allowing the SKA to constrain the dark matter particle mass and the heating level simultaneously.
- The first observationally informative result is now in hand. Analysing archival uGMRT observations of J352–15 — the brightest known radio-loud quasar in the EoR ($z = 5.82$) — at a sensitivity of $3.62 \text{ mJy beam}^{-1}$ per 6.1 kHz channel, Šoltinský et al. [30] obtain a null detection that nevertheless jointly constrains $\langle x_{\text{HI}} \rangle$ and the mean neutral-IGM temperature $\langle T_{\text{HI}} \rangle$: at 68% credibility, models with $\langle T_{\text{HI}} \rangle \lesssim 27 \text{ K}$ for $\langle x_{\text{HI}} \rangle = 0.1$ at $z \approx 5.6$ are disfavoured. This independently indicates substantial pre-heating of the neutral IGM above the adiabatic cooling floor, in the same direction as the HERA T_{S} limits (Section 2.6) but at much lower redshift and through a completely different systematic chain.

2.8 The duration of reionization from the kinetic Sunyaev–Zel’dovich effect

Free electrons in bulk motion during a *patchy* reionization Doppler-shift CMB photons, producing an excess of small-scale temperature anisotropy. Unlike τ , which is an integral of $n_{\text{e}}(z)$ (Eq. 1), the patchy kSZ power at $\ell \simeq 3000$ scales with the *duration* of the transition, and is therefore the only probe in this corpus that constrains Δz_{re} without assuming a source model.

- SPT-SZ + SPTpol report the first $\geq 3\sigma$ measurement of kSZ power, $D_{3000}^{\text{kSZ}} = 3.0 \pm 1.0 \mu\text{K}^2$, and, combined with their tSZ bispectrum, infer $\Delta z_{\text{re}} = 1.0^{+1.6}_{-0.7}$ ($\Delta z_{\text{re}} < 4.1$ at 95% confidence) [31].
- SPT-3G D1 measures $D_{3000}^{\text{kSZ}} = 1.98 \pm 0.87 \mu\text{K}^2$ and $D_{3000}^{\text{tSZ}} = 4.98 \pm 0.35 \mu\text{K}^2$ at 143 GHz (with the angular dependences of the SZ and CIB spectra free), and derives a 95% limit on the duration between ionized fractions of 25% and 75% of $\Delta^{50} z_{\text{re}} < 4.2$ (semi-analytic model), or between 5% and 95% of $\Delta^{90} z_{\text{re}} < 7.0$ (AMBER simulations) [2].

Caveat, stated by the authors themselves: the inferred kSZ power depends on the modelling of the tSZ–CIB correlation and on the assumed angular dependence of the SZ spectra, and the reionization inference depends further on the assumed level of homogeneous (late-time) kSZ power [2]. These limits are nevertheless consistent with, and independent of, the $\Delta z_{\text{re}} < 1.8$ (95% credibility) obtained by Sims et al. [21] from the 21-cm/Lyman-line/CMB joint fit — a genuine cross-check between two unrelated data sets.

2.9 Topology: reionization is patchy and inside-out

- Excess Ly α opacity scatter beyond density fluctuations [13, 14] and ultra-long dark gaps [15] require large-scale fluctuations in λ_{mfp} , in the UV background, and/or in $\langle x_{\text{HI}} \rangle$.
- JWST damping-wing analyses give ionized bubble radii evolving from $\log(R_{\text{b}}/\text{cMpc}) = 1.67^{+0.14}_{-0.16}$ at $z \simeq 7.1$ to $-0.69^{+0.89}_{-0.24}$ at $z \simeq 9.9$ [19]. These are up to a few dex larger than the cosmic average expected analytically at the same $\langle x_{\text{HI}} \rangle$, but are comparable to merged bubbles around bright galaxies in simulations — a selection effect that must be corrected before quoting a global $\langle x_{\text{HI}} \rangle$.
- Different reionization source models produce distinct bubble size distributions and hence distinct large-scale slopes of the 21-cm power spectrum, which is the primary route by which upcoming 21-cm data will identify the dominant agent [32].

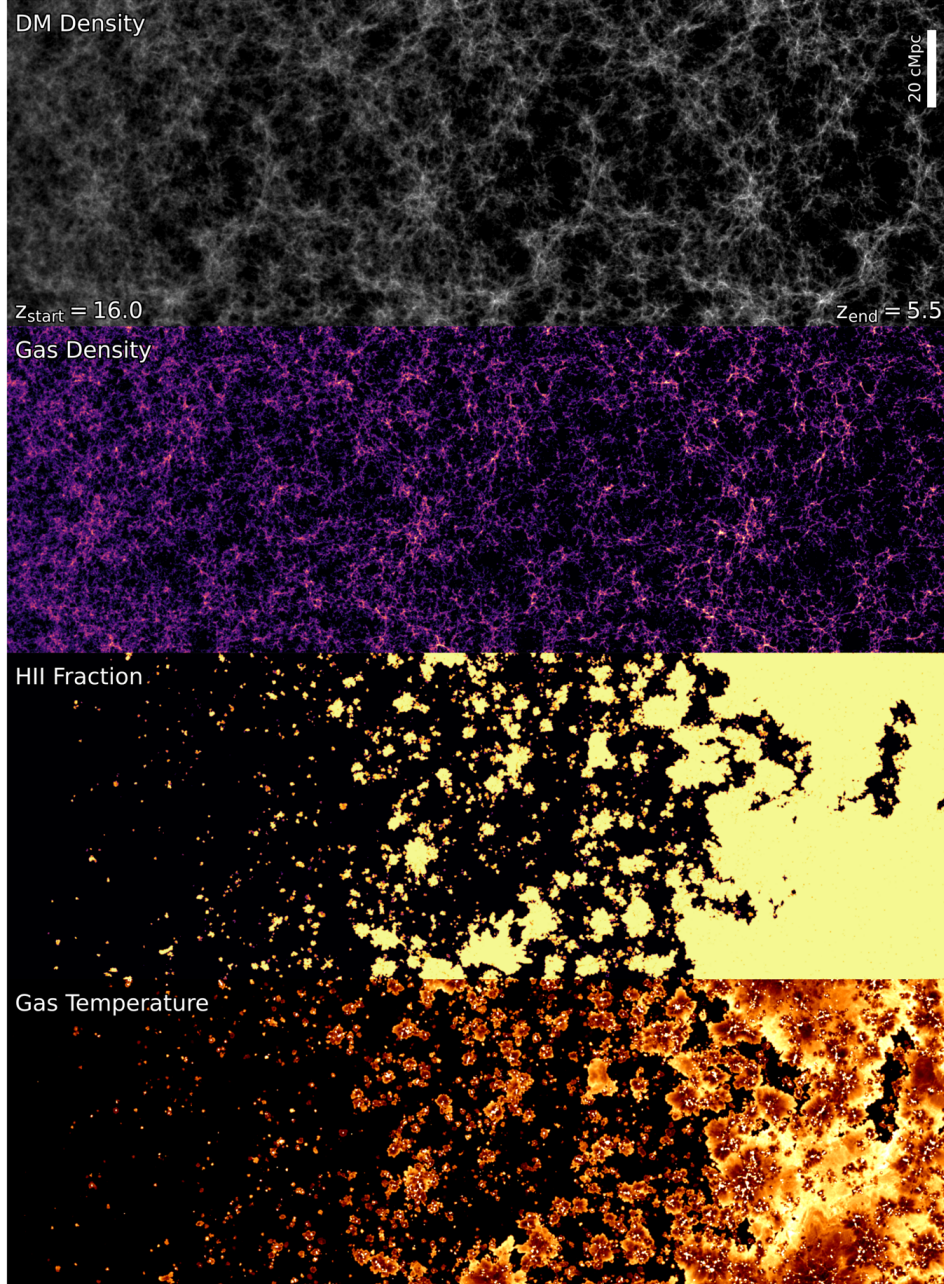


Figure 6: **Reionization is patchy and inside-out.** Mock light cones through the THESAN simulation from $z_{\text{start}} = 16.0$ (left) to $z_{\text{end}} = 5.5$ (right), showing dark matter density, gas density, H II fraction and gas temperature. The ionized regions (third panel) nucleate around the densest structures, grow, and merge: this is the geometry behind the opacity fluctuations of Section 2.2, the bubble-size measurements of Umeda et al. [19], and the selection bias that makes bubbles around bright galaxies unrepresentative. The bottom panel shows that photoheating tracks ionization exactly, which is why T_0 measures the ionized phase only.

Reproduced from Kannan et al. [32], Figure 2.

2.10 Helium reionization (a separate, later event)

He II requires ≥ 54.4 eV photons. Helium reionization is an AGN-driven event, distinct from and much later than hydrogen reionization — but the usual shortcut for *why* (“stars do not make such photons”) is too strong, and is contradicted by evidence in this corpus. The correct statement is that stars do not supply *enough* of them, and cannot supply them where they are needed:

- Metal-free stars make a non-negligible number. Murphy et al. [33], Table 4, give $N_{\text{HeII}}/N_{\text{H}} = 1.8 \times 10^{-2}$ for a top-heavy Pop III IMF and 9.6×10^{-3} for a Salpeter slope — one to two per cent, not zero. Rotating Pop III models reaching $T_{\text{eff}} \sim 2 \times 10^5$ K harden this further [34].
- It is observed. Strong He II $\lambda 1640$ emission — which requires He II recombination and hence > 54.4 eV photons absorbed locally — is detected at $\text{EW} = 21 \pm 4 \text{ \AA}$ in a sub- $L_{\star}$ galaxy at $z = 8.16$ [35], and at $\text{EW} > 20 \text{ \AA}$ with *no metal lines* at $z = 10.6$ near GN-z11, where Maiolino et al. [36] argue Pop III stars are the only plausible explanation.
- The simulation most often cited for the strong claim assumes it away. Eide et al. [27] conclude black holes “are the only sources which can fully ionize helium”, but state that they “do not account for contributions from stars that have evolved into binary systems, i.e. we use the ‘single star’ version of BPASS” — the choice that minimises hard-photon output, and precisely the one Rosdahl et al. [3] show matters at the factor-of-three level for ionizing escape.
- X-ray binaries emit ≥ 54.4 eV photons in quantity, yet still fail to reionize helium — for a different reason: their photoabsorbed SEDs are “too hard to produce significant He III regions around star-forming galaxies” [37], i.e. the photons free-stream instead of being absorbed.

The AGN conclusion survives all of this, because it rests on the global budget and the observed timing rather than on stars producing literally nothing:

- Observationally, He II reionization completes by $z \sim 3.0$ with a second IGM temperature peak $T_0 \sim 1.4 \times 10^4$ K [26].
- AGN with the measured UV luminosity function can reionize He II by $z = 2.9$ [38].
- In simulations, nuclear black holes are the *only* sources able to fully ionize helium, raising local temperatures by $\sim 10^4$ K [27].
- An AGN-dominated *hydrogen* reionization is disfavoured partly *because* it triggers He II reionization too early and overheats the IGM at $z \lesssim 5$ [39].

2.11 Consolidated parameter summary

Table 3 gathers the headline numbers of this section in one place; Section 4 then breaks each one down into the individual determinations behind it.

Table 3: Fiducial EoR parameters supported by observations, with the corpus source for each. All values as published.

Quantity	Value	Source
CMB optical depth τ	0.054 ± 0.007 (68% CL)	Planck Collaboration et al. [5]
τ (HFI/SRoll2)	$0.0566^{+0.0053}_{-0.0062}$ (68% CL)	Pagano et al. [9]
τ (astrophysics-anchored)	0.067 ± 0.011	Montero-Camacho and Mao [10]
Mid-point z_{50}	$7.16^{+0.15}_{-0.12}$; range $6.9 < z < 7.6$	Fan et al. [20], Sims et al. [21]
Duration	$\Delta z_{\text{re}} < 1.8$ (95% cred.)	Sims et al. [21]
End of reionization	$z \simeq 5.2\text{--}5.3$	Gaikwad et al. [7], Bosman et al. [14], Zhu et al. [15]
$\langle x_{\text{HI}} \rangle$ at $z = 6$	$0.174^{+0.093}_{-0.109}$	Gaikwad et al. [7]
$\langle x_{\text{HI}} \rangle$ at $z \simeq 7$	0.48 ± 0.26 to $0.59^{+0.11}_{-0.15}$	Mason et al. [17], Davies et al. [18]
$\langle x_{\text{HI}} \rangle$ at $z \simeq 10$	$0.92^{+0.08}_{-0.10}$	Umeda et al. [19]
$\langle \Gamma_{\text{HI}} \rangle$ at $z = 6$	$0.145^{+0.157}_{-0.087} \times 10^{-12} \text{ s}^{-1}$	Gaikwad et al. [7]
λ_{mfp} at $z = 6$	$0.75^{+0.65}_{-0.45} \text{ pMpc}$ (68%)	Becker et al. [24]
$\dot{n}_{\text{ion}}$ at $5 < z < 6$	$\simeq 0.6 \times 10^{51} \text{ s}^{-1} \text{ cMpc}^{-3}$	Gaikwad et al. [7]

continued on the next page

Table 3 continued from the previous page

Quantity	Value	Source
T_0 (ionized IGM), $z = 5.4\text{--}5.8$	$(1.05\text{--}1.2) \times 10^4 \text{ K}$	Gaikwad et al. [25]
T_S (neutral IGM), $z = 7.9$	$> 11.0 \text{ K}$ (95% cred.)	HERA Collaboration et al. [1]
T_S (neutral IGM), $z = 10.4$	$> 6.2 \text{ K}$ (95% cred.)	HERA Collaboration et al. [1]
$L_{X<2\text{ keV}}/\text{SFR}$	$10^{39.9\text{--}41.6} \text{ erg s}^{-1} (\text{M}_\odot \text{ yr}^{-1})^{-1}$ (68%)	HERA Collaboration et al. [1]
He II reionization end	$z \sim 3.0$, $T_0 \sim 1.4 \times 10^4 \text{ K}$	Villasenor et al. [26]
Quasar lifetime t_Q	$\lesssim 7 \text{ Myr}$	Đurovčiková et al. [16]
Duration Δz_{re} (kSZ)	$1.0^{+1.6}_{-0.7}$; < 4.1 (95% conf.)	Reichardt et al. [31]
Duration $\Delta^{50}_{z_{\text{re}}} \text{ (kSZ)}$	< 4.2 (95%)	Chaubal et al. [2]
$\langle T_{\text{HI}} \rangle$ (neutral IGM), $z \approx 5.6$	$\gtrsim 27 \text{ K}$ for $\langle x_{\text{HI}} \rangle = 0.1$ (68% cred.)	Šoltinský et al. [30]
Clumping factor C_{HII}	≈ 12 (observational); $6.2^{+4.1}_{-2.1}$ required	Davies et al. [40], Austin et al. [41]

3 The agents of reionization and heating

3.1 The criteria the literature uses

This subsection answers the explicit request to state the criteria used to compute the contribution of a population of agents to the ionization and heating of the IGM. There are five, and they are logically distinct.

3.1.1 Criterion 1 — the global ionizing photon budget

The near-universal starting point [e.g. 8] is the volume-filling-factor ODE

$$\dot{Q}_{\text{HII}} = \frac{\dot{n}_{\text{ion}}}{\langle n_{\text{H}} \rangle} - \frac{Q_{\text{HII}}}{t_{\text{rec}}}, \quad (3)$$

with the ionizing emissivity of a stellar population

$$\dot{n}_{\text{ion}} = f_{\text{esc}} \xi_{\text{ion}} \rho_{\text{UV}} \quad (\text{or } \rho_{\text{SFR}}), \quad (4)$$

and the IGM recombination time

$$t_{\text{rec}} = \left[C_{\text{HII}} \alpha_{\text{B}}(T) (1 + Y_{\text{p}}/4X_{\text{p}}) \langle n_{\text{H}} \rangle (1 + z)^3 \right]^{-1}. \quad (5)$$

What Q_{HII} is. Q_{HII} is the *volume filling factor of ionized hydrogen*: the fraction of the total comoving volume of the Universe that sits inside an H II region. It is a geometric quantity, not a chemical one. The picture behind it is the one shown in Figure 6: reionization does not proceed by the whole IGM ionizing a little at a time, but by discrete bubbles opening around individual galaxies, growing, and eventually merging until they fill space. Q_{HII} counts the fraction of space those bubbles have swallowed — 0 before the first sources light up, 1 at overlap. The model is explicitly *two-phase*: gas is idealised as either fully ionized (inside a bubble) or fully neutral (outside one), with no intermediate state and no structure within either phase. Equation 3 is then just bookkeeping on that volume. Its left-hand side is the rate at which ionized volume grows; the first term on the right is the rate at which sources supply the photons to grow it, and the second is the rate at which recombinations give volume back. Both terms have dimensions of inverse time (Section 1.4), and reionization completes when the source term has outrun the sink term for long enough.

Each term, and where its value comes from. The **source term** $\dot{n}_{\text{ion}}/\langle n_{\text{H}} \rangle$ divides an emission rate per unit comoving volume by a number density per unit comoving volume, which converts “ionizing photons emitted per second per cMpc³” into “hydrogen atoms ionized per second per hydrogen atom” — i.e. the fraction of the volume newly ionized per unit time, on the assumption that each escaping photon ionizes exactly one atom. $\dot{n}_{\text{ion}}$ itself is either *measured* directly from the Ly α forest ($\simeq 0.6\text{--}0.7 \times 10^{51} \text{ s}^{-1} \text{ cMpc}^{-3}$ at $4.9 \leq z \leq 6.0$; Table 2 and Section 2.4) or *built* from Eq. 4, in which case its three factors are sourced independently: ρ_{UV} by integrating the observed UV luminosity function down to a cut-off $M_{\text{UV}}^{\text{lim}}$ (HST and JWST imaging; the faint-end slope and the cut-off are both uncertain), ξ_{ion} from H α /UV ratios or SED fitting of the same galaxies, and f_{esc} — the one factor that cannot be observed at all at $z > 6$, because the intervening IGM is opaque to exactly the photons one would need to detect [22]. The denominator $\langle n_{\text{H}} \rangle$ is *not* a free parameter: it is fixed by cosmology to $1.895 \times 10^{-7} \text{ cm}^{-3}$ comoving from $\Omega_{\text{b}} h^2$ and Y_{p} [5], as computed in Section 1.4. The **sink term** $Q_{\text{HII}}/t_{\text{rec}}$ carries the physics that makes reionization hard. Within t_{rec} (Eq. 5), α_{B} is the case-B recombination coefficient — “case B” because recombinations straight to the ground state release another ionizing photon and so do not count as a net loss — and it depends on temperature, which is measured from the Ly α forest (Section 2.5, $T_0 \simeq 1.1 \times 10^4 \text{ K}$). $C_{\text{HII}} = \langle n_{\text{H}}^2 \rangle / \langle n_{\text{H}} \rangle^2$ is the clumping factor, which corrects for the fact that recombination is a two-body process and therefore runs fastest in the densest gas: a clumpy IGM recombines faster than a smooth one of the same mean density. It is the least settled quantity in the whole equation and has its own discussion in Section 3.1.2. The factor $(1 + Y_{\text{p}}/4X_{\text{p}}) = 1.081$ converts hydrogen density into *electron* density: helium is singly ionized wherever hydrogen is, because stellar spectra remain strong between 13.6 and 54.4 eV (Figure 8), so each helium nucleus donates one electron per four units of mass. It would become $(1 + Y_{\text{p}}/2X_{\text{p}})$ only after He II reionization, which does not happen until $z \sim 3$. Finally $(1 + z)^3$ converts the comoving density to a proper one, since recombination responds to the physical density of the gas.

The pitfall: Q_{HII} is not what the observations return. It is tempting — and very common — to read Q_{HII} off Eq. 3 and compare it directly with the $1 - \langle x_{\text{HI}} \rangle$ of Table 1. The two are related but not identical. Decomposing the volume-averaged neutral fraction into the two phases,

$$\langle x_{\text{HI}} \rangle = (1 - Q_{\text{HII}}) + Q_{\text{HII}} \langle x_{\text{HI}} \rangle_{\text{ionized}}, \quad (6)$$

so that $1 - \langle x_{\text{HI}} \rangle = Q_{\text{HII}} (1 - \langle x_{\text{HI}} \rangle_{\text{ionized}})$. Since the residual neutral fraction inside an ionized bubble is $\langle x_{\text{HI}} \rangle_{\text{ionized}} \sim 10^{-4}\text{--}10^{-5}$, the approximation $Q_{\text{HII}} \simeq 1 - \langle x_{\text{HI}} \rangle$ is excellent *while reionization is in progress*, when the neutral phase dominates the neutral budget. It fails in two places. First, after overlap: Q_{HII} saturates at 1 by construction and Eq. 3 stops evolving, whereas the measured $\langle x_{\text{HI}} \rangle$ keeps falling by almost four orders of magnitude, from 1.7×10^{-1} at $z = 6.0$ to 2.5×10^{-5} at $z = 4.9$ (Table 2). That entire late phase — where the mean free path, the opacity fluctuations and the residual Lyman-limit systems live (Sections 2.2 and 2.4) — is invisible to the filling factor. Second, the two quantities are weighted differently by different probes: Eq. 3 tracks a *volume* average, while recombinations (and hence C_{HII}) are set by a *mass*-weighted quantity, and an individual damping-wing sightline samples neither cleanly — bright galaxies sit in atypically large bubbles [19]. When comparing a model $Q_{\text{HII}}(z)$ against the data compilation of Figure 1, these are the assumptions being made silently.

The contribution of a population is then the ratio of its $\dot{n}_{\text{ion}}$ to the total, and the population is judged viable if Eq. 3 reproduces the observed $\langle x_{\text{HI}} \rangle(z)$ and τ . Four quantities carry essentially all of the systematic uncertainty:

f_{esc} : not directly measurable at $z > 6$ because the IGM is opaque [22]; must be inferred from simulations, from low- z analogues, or from nebular-line proxies.

ξ_{ion} : the canonical pre-JWST value was $\log_{10} \xi_{\text{ion}} / (\text{Hz erg}^{-1}) \approx 25.2$ [as used by 8, 42]; JWST now finds $\approx 25.5\text{--}26.0$ [42], with $\log_{10} \xi_{\text{ion}} = 25.80 \pm 0.14$ for ultra-faint lensed galaxies [43]. This single change is worth a factor ~ 4 in the budget.

$M_{\text{UV}}^{\text{lim}}$: extrapolating the UV LF from -15 to -11 changes the answer materially [42].

C_{HII} : $C_{\text{HII}} \simeq 3$ is the common fiducial [8]; my cross-check gives $t_{\text{rec}}(z=7) = 704 \text{ Myr}$ for $C_{\text{HII}} = 3$ versus 2113 Myr for $C_{\text{HII}} = 1$ (at $T = 2 \times 10^4 \text{ K}$, the temperature Robertson et al. [8] adopt, where $\alpha_{\text{B}} = 1.43 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$) — i.e. a factor 3 in the recombination sink. **The fiducial value is now seriously contested** (see Section 3.1.2).

3.1.2 The clumping factor and small-scale structure: a first-class systematic

C_{HII} is not a nuisance constant but a physically evolving quantity, and the choice made for it moves the budget verdict by a factor ~ 2 .

- **Physics.** The IGM is expected to clump on scales down to the $\sim 10^4 M_{\odot}$ Jeans mass of unheated gas. Radiation-hydrodynamic simulations resolving that scale show the clumping factor of ionized gas *peaks at 5–20 about $\Delta t = 10 \text{ Myr}$ after the ionization front passes*, depending on the incident intensity, redshift, and how much the gas had been pre-heated by early X-ray sources, and relaxes to ≈ 3 only by $\Delta t = 300 \text{ Myr}$ [44]. Because the relaxation is slow, un-relaxed structure boosts the total number of recombinations by $\approx 50\%$ and drives spatial fluctuations in λ_{mfp} that persist for several hundred Myr *after* reionization completes — one of the physical origins of the opacity fluctuations of Section 2.2.
- **Observational estimate.** Davies et al. [40] connect C_{HII} to the mean free path by imposing ionization equilibrium in the ionized phase, and find that the measured λ_{mfp} and $\langle \Gamma_{\text{HI}} \rangle$ at $z = 5\text{--}6$ (Table 2) imply a *global* $C \approx 12$ — far above the $C \approx 3$ obtained from radiation-hydrodynamic simulations of the reionization process. At $C \approx 12$, galaxies must supply roughly twice as many ionizing photons, ≈ 3 photons per baryon, to reionize by $z \sim 6$.
- **Consequence for the budget crisis.** This cuts the opposite way from the JWST ξ_{ion} excess, and can absorb it. Austin et al. [41], from 1721 galaxies at $5.6 < z < 6.5$ across $\sim 550 \text{ arcmin}^2$ of JWST/NIRCam imaging (NEP, JADES, PRIMER), measure a *magnitude-independent* efficiency $\log_{10}(\xi_{\text{ion}}/\text{Hz erg}^{-1}) = -0.006_{-0.017}^{+0.019} M_{\text{UV}} + 25.05_{-0.34}^{+0.39}$ — consistent with the canonical HST value, not with the elevated JWST ones — and an escaping production rate $\log_{10}(\dot{n}_{\text{ion}}/\text{s}^{-1} \text{ Mpc}^{-3}) = 50.31_{-0.06}^{+0.07}$ for $M_{\text{UV}} < -17$. Extrapolating to $M_{\text{UV,lim}} = -13.5$, they find that a recombination-weighted $C_{\text{HII,rec}} = 6.2_{-2.1}^{+4.1}$ is *required* for stable reionization at $z \simeq 6$, and that a clumping factor of this magnitude resolves the photon budget crisis outright.

Consistency note. My cross-check (Section 1.4) gives 3.4 ionizing photons per H atom per Gyr for the measured emissivity. The ≈ 3 photons per baryon required at $C \approx 12$ [40] is the same number viewed as a total rather than a rate, so the two statements are mutually consistent; they are *not* independent confirmations of each other, since both rest on the same Gaikwad et al. [7]/Becker et al. [24] measurements.

3.1.3 Criterion 2 — matching the *measured* ionizing background

A budget that integrates correctly can still fail locally. The stronger test is to require that a source model reproduce simultaneously the measured $\langle \Gamma_{\text{HI}} \rangle(z)$, $\lambda_{\text{mfp}}(z)$ and the *distribution* (not just the mean) of Ly α effective optical depths. This is the criterion applied by Gaikwad et al. [7], Asthana et al. [39] and Bosman et al. [14], and it is what rules out the AGN-only scenario: an AGN-only model calibrated to the mean transmission is inconsistent with the observed τ_{eff} *distribution* [39].

3.1.4 Criterion 3 — energy deposition and the heating fraction

For heating, the relevant question is not how many photons a source produces but *what fraction of the injected energy ends up as heat*. A photon of energy E photoionizes an atom and deposits $E - E_i$ into a fast primary electron, which then partitions its energy between further collisional ionization, collisional excitation (producing Ly α photons) and Coulomb heating. The canonical

calculation is the Monte Carlo of Furlanetto and Stoeve [45], plotted in Figure 7, which gives f_{heat} , f_{ion} and f_{exc} as functions of the electron energy E and the background ionized fraction x_e :

$$\bar{f}_{\text{heat}} = \frac{\sum_i \int_{E_{\min}}^{\infty} dE_{\gamma} E_{\gamma}^{-\alpha-1} n_i \sigma_i(E_{\gamma}) (E_{\gamma} - E_i) f_{\text{heat}}}{\sum_i \int_{E_{\min}}^{\infty} dE_{\gamma} E_{\gamma}^{-\alpha} n_i \sigma_i(E_{\gamma})}, \quad (7)$$

Here the source spectrum is $L_{\nu} \propto \nu^{-\alpha}$, so the photon *number* per unit energy goes as $E_{\gamma}^{-\alpha-1}$; E_i is the ionization threshold of species i ; σ_i are the photoionization cross-sections of Verner et al. (1996); and $E_{\min}$ is the attenuation threshold, below which photons are absorbed before reaching the point in question, so the integrals start there rather than at E_i .

The asymmetry between numerator and denominator is deliberate and is the whole content of the normalisation. The numerator carries $E_{\gamma}^{-\alpha-1}$ and a factor $(E_{\gamma} - E_i)$ — the energy handed to the *secondary* electron, of which a fraction f_{heat} becomes heat. The denominator carries $E_{\gamma}^{-\alpha}$ and *no* such factor, so it is the total photon energy *absorbed*, including the E_i spent on the initial ionization. Consequently $\bar{f}_{\text{heat}} + \bar{f}_{\text{ion}} + \bar{f}_{\text{Ly}\alpha}$ is *much smaller than unity at low energies*, as Furlanetto and Stoeve [45] note explicitly: the energy invested in the primary ionization is never recovered by the secondary. Normalising instead to $(E_{\gamma} - E_i)$ would make the three fractions sum to one and would erase exactly that physical point.

The physically important results are: (i) the deposition fractions depend strongly on x_e and E but are *nearly independent of the gas density*; (ii) electrons with $E < 10.2 \text{ eV}$ cannot excite or ionize hydrogen, so *all* their energy becomes heat; (iii) at $E \sim 1\text{--}10 \text{ keV}$ the energy splits roughly equally between ionization, heating and excitation, and does *not* continue to favour ionization at higher energies (contrary to the naive Bethe expectation), because each ionization creates a low-energy secondary that thermalises [45].

3.1.5 Criterion 4 — the mean free path / SED-hardness criterion

Because the photoionization cross-section falls steeply with energy ($\sigma \propto E^{-3}$ above threshold), the comoving mean free path of an X-ray photon rises steeply with energy. Consequently:

- Soft X-rays ($\lesssim 1 \text{ keV}$) are absorbed close to their sources and produce *clustered, inhomogeneous* heating.
- Hard X-rays ($\gtrsim 2 \text{ keV}$) free-stream and produce nearly *uniform* heating, or escape entirely and contribute to the unresolved X-ray background instead.

This is why the SED shape, not just the bolometric luminosity, is a criterion in its own right: Pacucci et al. [46] show that if the soft band is dominated by hot-ISM thermal emission rather than by harder X-ray binaries, the large-scale 21-cm power is boosted by roughly a factor of three, and that the *redshift of the peak* of the large-scale power constrains the X-ray luminosity and host halo mass while the *amplitude* constrains the soft-band SED, breaking the degeneracy with bolometric luminosity. Gessey-Jones et al. [47] make the same argument for cosmic rays, whose heating is short-ranged and therefore leaves more small-scale power than X-ray heating.

What the spectra actually look like. Figure 8 makes the criterion concrete: it plots the mean spectrum of each agent against photon energy, from the H I edge to 2 keV, with the three ionization thresholds marked. The four curves occupy almost disjoint parts of the spectrum, which is the whole reason the ionizing and heating budgets decouple.

- **Stars** dominate everything by three to seven orders of magnitude between 13.6 and 54.4 eV — and then fall off a cliff. In the words of Eide et al. [27], the volume-averaged galaxy spectra “are strongest at H I and He I ionizing frequencies, and fall by orders of magnitude near the He II ionization threshold”. A normal stellar population is therefore an almost

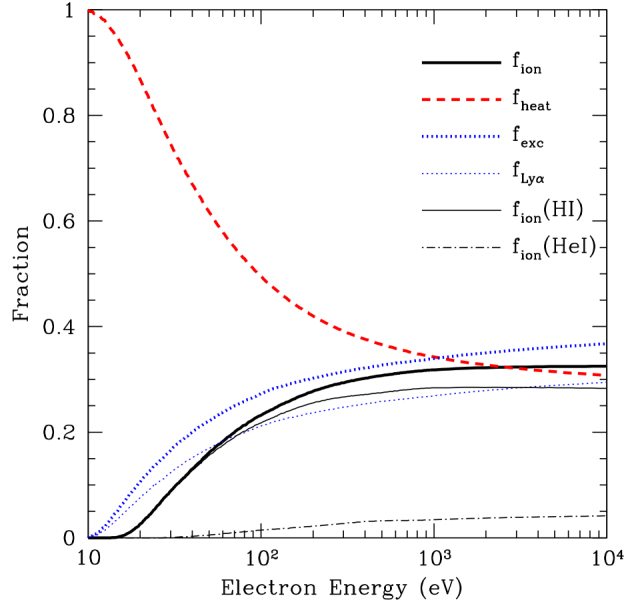


Figure 7: **Why a hard photon is not simply an ionizing photon.** Fractions of a fast electron’s initial energy deposited as ionization (f_{ion}), heat (f_{heat}) and line excitation (f_{exc} , with the Ly α part shown separately), for an IGM with $x_e = 0.01$. Below 10.2 eV an electron cannot excite or ionize hydrogen, so *all* its energy becomes heat. Above ~ 1 keV the three channels split roughly equally and stay flat — contrary to the naive Bethe expectation — because each ionization creates a low-energy secondary that thermalises. These are the f_{heat} , f_{ion} entering Eq. 7, and no statement about the relative ionizing versus heating power of X-rays or cosmic rays is meaningful without them.

Reproduced from Furlanetto and Stoever [45], Figure 1.

monochromatic source on this scale: it supplies the hydrogen budget and essentially nothing else.

- **Nuclear black holes** follow a power law all the way to keV energies, which is why they are the only population that reionizes helium (Section 3) despite being far too rare to matter for hydrogen.
- **Hot ISM** emission is flat to ~ 200 eV and then declines; **X-ray binaries** do the opposite, rising to a plateau above ~ 500 eV. Below ~ 1 keV the two are comparable, which is exactly the ambiguity Pacucci et al. [46] exploit — see Figure 9.

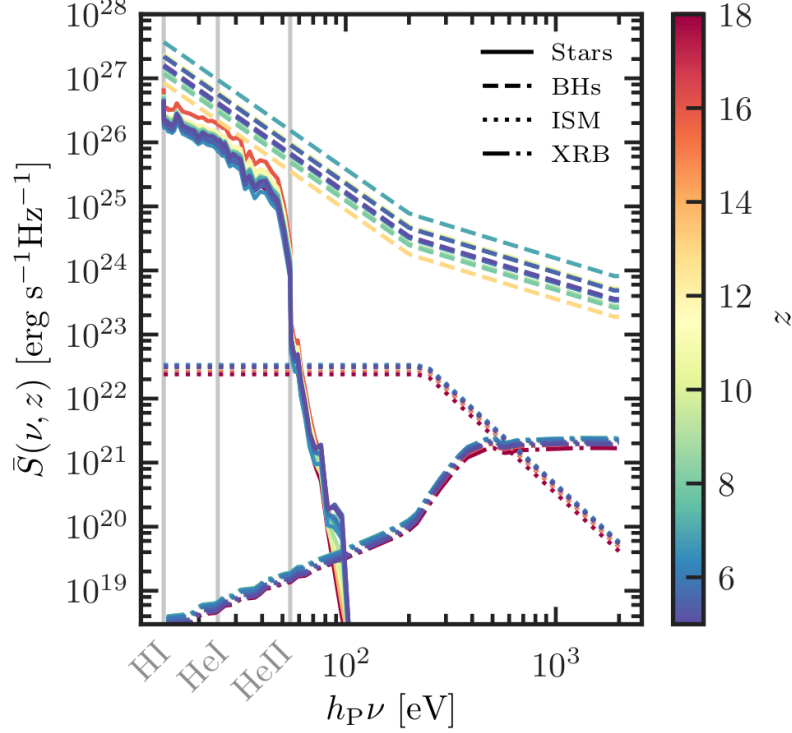


Figure 8: **Where each agent puts its photons.** Average spectral energy distribution of stars (solid), nuclear black holes (dashed), hot ISM (dotted) and X-ray binaries (dash-dot-dotted) as a function of photon energy, coloured by redshift; grey vertical lines mark the H I, He I and He II thresholds. Stellar spectra are from BPASS. Note the near-vertical stellar drop at 54.4 eV: it is the single most important feature on this plot, because it is why hydrogen and helium reionization are separate events driven by different populations.

Reproduced from Eide et al. [27], Figure 1.

Relative photon abundance by energy band. For a source with $\epsilon_\nu \propto \nu^{-\alpha}$ the photon emission rate per unit frequency is $\dot{n}_\nu = \epsilon_\nu / h\nu \propto \nu^{-\alpha-1}$, so the fraction of hydrogen-ionizing photons emitted above an energy E is just $(E/E_{\text{HI}})^{-\alpha}$. Table 4 evaluates this for the effective ionizing-source index $\alpha_s = 2.0 \pm 0.6$ measured by Gaikwad et al. [7] and for the AGN power law $\alpha_{\text{AGN}} = 1.7$ adopted by Asthana et al. [39] (who print it as $\alpha = -1.7$ in the $\nu^{+\alpha}$ convention; this document uses $\epsilon_\nu \propto \nu^{-\alpha}$ throughout). **These percentages are my own calculation** from those published indices (script: `photon_energy_fractions.py`; the closed form is verified with `sympy`), not values quoted in either paper.

[Important caveat, and the reason Figure 8 matters more than Table 4: a single power law is the wrong shape for a stellar population above the He II edge. α_s is an *effective* index fitted so that the escaping emissivity reproduces the Ly α forest over a narrow range above 13.6 eV; extrapolating it to 54.4 eV

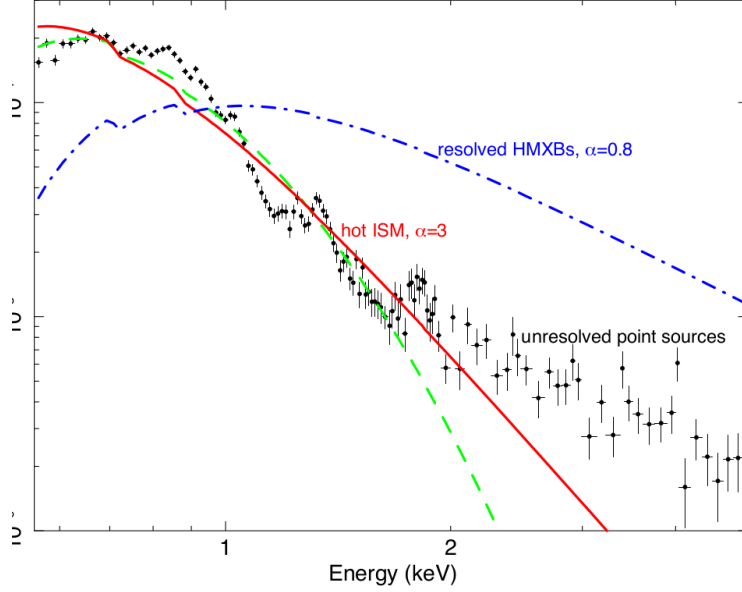


Figure 9: **The soft X-ray band, measured rather than assumed.** Composite *observed* X-ray SED of 21 local star-forming galaxies, in photons $\text{cm}^{-2} \text{s}^{-1} \text{keV}^{-1}$. Below $\sim 2 \text{ keV}$ the emission is dominated by the hot ISM (red solid, a power law of energy index $\alpha = 3$; green dashed, a thermal bremsstrahlung fit); above $\sim 2 \text{ keV}$ by unresolved point sources, chiefly high-mass X-ray binaries (blue dot-dashed, $\alpha = 0.8$). All curves include intrinsic absorption by $N_{\text{H}} \sim 2\text{--}3 \times 10^{21} \text{ cm}^{-2}$. The two components have comparable soft-band luminosity per unit SFR but *dramatically different* spectral shapes — which is why the split between them changes the predicted 21-cm power by a factor of three.

Reproduced from Pacucci et al. [46], Figure 1.

Table 4: Share of hydrogen-ionizing photons by energy band for a power-law emissivity $\epsilon_\nu \propto \nu^{-\alpha}$. Thresholds: H I 13.598 eV, He I 24.587 eV, He II 54.418 eV.

α	source	13.6–24.6 eV	24.6–54.4 eV	> 54.4 eV
1.4	$\alpha_s - 1\sigma$	56.4%	29.3%	14.3%
2.0	α_s (fiducial)	69.4%	24.3%	6.2%
2.6	$\alpha_s + 1\sigma$	78.6%	18.7%	2.7%
1.7	α_{AGN}	63.5%	27.1%	9.5%

predicts $\sim 6\%$ of photons in the He II band, whereas the BPASS spectra in Figure 8 fall by several orders of magnitude there, putting the true stellar fraction far below 1%. Use the table for the AGN row and for the 13.6–24.6 eV split; use the figure for anything involving the He II edge.]

3.1.6 Criterion 5 — calibrated luminosity-to-SFR (or -to-mass) scalings

Finally, converting a source population into an energy injection rate requires an empirical scaling: L_X/SFR for X-ray binaries, the AGN UV luminosity function for quasars, a cosmic-ray escape efficiency for supernova remnants. These are calibrated locally and extrapolated, and the extrapolation is now testable: HERA excludes the local, metal-enriched L_X –SFR relation at $> 99\%$ of posterior volume, requiring instead low-metallicity HMXB models [1]. Dasgupta et al. [48] further show that the deterministic L_X –SFR scaling itself breaks down in high- z , low-SFR regions hosting few sources: sampling L_X stochastically from an X-ray luminosity function enhances 21-cm power at $k > 0.3 \text{ cMpc}^{-1}$ while leaving the global signal and large-scale power unchanged.

3.1.7 Classification convention used in this document

This threshold convention is mine, not a literature standard, adopted so that Table 5 is unambiguous. Referring to the total ionizing photon budget (or total energy injection, for heating): *dominant* $> 50\%$; *significant* 10–50%; *minor* 1–10%; *negligible* $< 1\%$. Where a paper states a percentage I use it; where it states only “dominant”/“negligible” I reproduce the paper’s word and say so.

3.2 Catalogue of agents

3.2.1 Star-forming galaxies: Lyman-continuum photons

Ionization: dominant. This is the consensus position of the corpus. Robertson et al. [8] showed that the HST UV LF plus $f_{\text{esc}} = 0.2$ and $\log_{10} \xi_{\text{ion}} = 53.14 [\text{s}^{-1}/(\text{M}_{\odot} \text{yr}^{-1})]$ simultaneously match $\tau = 0.066 \pm 0.012$ (the then-current *Planck* value) and the observed neutrality evolution over $6 \lesssim z \lesssim 10$, without requiring a significant $z \gg 10$ population. Radiative-transfer simulations reach the same conclusion mechanistically: “stars are the main driver of hydrogen reionization and consequently of the thermal history of the IGM” [27].

Which galaxies, though, is genuinely unsettled. Three positions coexist in the current literature:

Faint dwarfs dominate. Finkelstein et al. [23] find a solution with average $f_{\text{esc}} < 5\%$ throughout the bulk of the EoR provided galaxies form down to the atomic cooling limit and ξ_{ion} rises towards fainter magnitudes and higher z ; in this model $M_{\text{UV}} > -15$ galaxies dominate the emissivity. Atek et al. [43] measure $\log_{10} \xi_{\text{ion}}/(\text{Hz erg}^{-1}) = 25.80 \pm 0.14$ for eight lensed galaxies at $M_{\text{UV}} \sim -17$ to -15 (down to $0.005 L^*$) — a factor ~ 4 above the canonical value — and conclude that faint galaxies alone exceed the requirement even for $f_{\text{esc}} \sim 5\%$.

Bright “oligarchs” dominate. Naidu et al. [6] require $f_{\text{esc}} = 0.21^{+0.06}_{-0.04}$ for $M_{\text{UV}} < -13.5$ galaxies, propose $f_{\text{esc}} \propto \Sigma^{0.4 \pm 0.1}$ (star-formation surface density), and find that $< 5\%$ of galaxies with $M_{\text{UV}} < -18$ and $\log(M_{\star}/\text{M}_{\odot}) > 8$ supply $\gtrsim 80\%$ of the budget, with faint sources ($M_{\text{UV}} > -16$) necessarily relegated to a minor role in order to keep $\langle x_{\text{HI}} \rangle$ high at $z = 7$ –8.

A leaking minority dominates, and the balance evolves. The most recent direct constraint in the corpus [22] fits 1428 JWST/NIRSpec spectra at $5 < z < 10$ with a picket-fence model and finds that $\sim 20\%$ of sources — those with $f_{\text{esc}} > 10\%$ — produce $\sim 87\%$ of the ionizing photons; bright sources ($M_{\text{UV}} \sim -20$) dominate at $z = 5$ –6, whereas at $z > 6$ faint galaxies ($M_{\text{UV}} > -18$) contribute roughly equally.

Stellar physics matters at the factor level. Including binary stars in the stellar population synthesis raises f_{esc} to $\approx 7\text{--}10\%$, about $3\times$ the single-star value (the luminosity-weighted average escape fraction is 8.5% for binaries versus 2.7% for singles, a factor ≈ 3.1), because the ionizing luminosity declines more slowly with age; in the SPHINX simulations this is the difference between volumes that are 99.9% ionized by $z \approx 7$ and volumes that fail to reionize by $z = 6$ at all [3].

Heating: dominant for the ionized phase, negligible for the neutral phase. Stellar LyC photons have a short mean free path and deposit their excess energy inside H II regions, photoheating them to $\sim 1\text{--}2 \times 10^4$ K [25–27]. They do essentially nothing to the gas between H II regions, which is why the pre-reionization 21-cm signal is a clean probe of the non-stellar agents.

3.2.2 Population III stars

Ionization: negligible at $z \lesssim 10$; potentially dominant only at $z \gtrsim 30$. Pop III stars are individually far more efficient ionizers — harder spectra produce $\sim 10\times$ more ionizing photons than Pop II [49] — but the Pop III era is self-terminating. Kulkarni et al. [49] show with a chemically-tracking semi-analytic model that a single generation self-enriches its host above the critical metallicity $Z_{\text{crit}} = 10^{-4} Z_{\odot}$ in $\sim 10^6$ yr, so Pop III stars supply $\gtrsim 50\%$ of ionizing photons only at $z \gtrsim 30$ and $< 1\%$ at $z = 10$; comparison of the predicted abundance pattern with measured $z \lesssim 6$ DLA abundances bounds the high-mass Pop III fractional contribution to the ionization rate at $\lesssim 10\%$ at $z = 10$.

The per-star ionizing output is now computed from Geneva zero-metallicity models over $1.7\text{--}500 M_{\odot}$ [33]: rotation changes ionizing photon production by up to 25% , convective overshooting by $\sim 20\%$, and varying the IMF slope and mass limits by $20\text{--}30\%$ at fixed population mass. These are the systematic uncertainties on any Pop III budget.

Heating: indirect but possibly significant, via (i) Pop III supernova cosmic rays (below) and (ii) Pop III X-ray binaries [48]; and the HERA requirement of low-metallicity HMXB L_X –SFR scalings [1] points in the same direction.

3.2.3 Quasars and AGN

Hydrogen ionization: contested; most of the corpus says minor to significant, not dominant.

- Madau and Haardt [50] constructed the strongest AGN-only case: if the claimed abundant faint AGN population at $z = 4\text{--}6$ is real and its emissivity persists to $z \sim 10$, AGN can reionize hydrogen *and* helium with little stellar contribution, giving a flat $\langle \Gamma_{\text{HII}} \rangle$ over $2 < z < 5$, hydrogen fully reionized by $z = 5.7$ and $\tau = 0.056$. The model predicts helium doubly reionized before $z = 4$ and a 50% He III volume fraction already when hydrogen completes; He contributes $\sim 13\%$ of τ .
- Kulkarni et al. [38], from a homogenised sample of $> 80,000$ colour-selected AGN from $z = 0$ to 7.5 , find the AGN contribution to the hydrogen photoionization rate at $z \sim 6$ to be *sub-dominant*, $< 3\%$, rising to $\sim 10\%$ only if the LF is extrapolated to $M_{1450} = -18$.
- Dayal et al. [51], with a semi-analytic model of galaxies and black holes, find AGN contribute only $10\text{--}25\%$ of the *cumulative* ionizing emissivity by $z = 4$ in models that match the reionization constraints, with stellar radiation from $M_{\star} < 10^9 M_{\odot}$ galaxies dominating at $z > 6$.
- Asthana et al. [39] post-process a Sherwood-Relics simulation with radiative transfer and find a “QSO-assisted” model (AGN supplying 17% of the hydrogen-ionizing emissivity) fits the observed Ly α optical depth distribution, whereas a QSO-only model does not, and overheats the IGM at $z \lesssim 5$ through premature He II reionization.
- **Dissenting, recent:** Singha et al. [52] build a unified rest-UV AGN LF separating Type I/Type II and treating “little red dots” and X-ray-selected sources as magnitude-filtered subsets of

Type I, and anchor the LyC escape fraction to outflow incidence rather than assuming quasar-like values. Integrating over $-27 < M_{\text{UV}} < -17$ they obtain $\dot{N}_{\text{ion,AGN}} = (3.77^{+1.08}_{-0.95}) \times 10^{51} \text{ s}^{-1} \text{ cMpc}^{-3}$ — nearly twice earlier estimates — and conclude AGN supply 31–75% of ionizing photons at $z > 5$ for $f_{\text{esc}}^{\text{gal}} = 0.03\text{--}0.20$, with $\Gamma_{\text{HI}} \sim (0.5\text{--}2) \times 10^{-12} \text{ s}^{-1}$ at $z \sim 5\text{--}6$. [Note: their quoted $\dot{N}_{\text{ion,AGN}}$ exceeds the total measured emissivity $\simeq 0.6 \times 10^{51} \text{ s}^{-1} \text{ cMpc}^{-3}$ of Gaikwad et al. [7] by a factor ~ 6 . I have not reconciled this; it may reflect a different definition (emitted vs. escaping photons) or a genuine inconsistency. Do not quote both numbers side by side without checking the definitions in the papers.]

Helium ionization: dominant (Sec. 2.8): AGN are the only population that supplies $\geq 54.4 \text{ eV}$ photons in quantity [27, 38, 50].

Heating: dominant for the post-reionization IGM, negligible for the bulk neutral IGM. Nuclear black holes are rare, so they are negligible for hydrogen reionization, but they raise local temperatures by $\sim 10^4 \text{ K}$ where they do ionize helium [27]; heat input from helium reionization dominates the thermal balance of the IGM *after* hydrogen reionization [50], consistent with the second T_0 peak measured by Villasenor et al. [26].

3.2.4 High-mass X-ray binaries

Ionization: negligible. Madau and Fragos [37] follow compact-binary evolution with population synthesis calibrated to $z < 4$ galaxies and integrate the rate and energy equations: the XRB contribution to ionization of the bulk IGM never exceeds an electron fraction of 1%.

Heating: the leading candidate, and now observationally implicated.

- The filtered XRB background peaks between 1 and 2 keV; direct He I photoionization is the main source of IGM heating; the largely neutral gas between H II cavities rises above T_{CMB} only at $z < 10$, once the H II filling factor already exceeds 0.1 [37]. This makes 21-cm *emission* a late phenomenon in this scenario.
- HERA’s T_{S} lower limits establish that the IGM was heated above the adiabatic cooling limit at least as early as $z = 10.4$, and the inferred $L_{X<2\text{keV}}/\text{SFR}$ excludes the local relation at $> 99\%$ of posterior volume, requiring evolved low-metallicity stellar populations [1].
- The deterministic $L_X\text{--SFR}$ scaling is itself an approximation: stochastic sampling from an X-ray LF enhances small-scale ($k > 0.3 \text{ cMpc}^{-1}$) 21-cm power while leaving the global signal unchanged [48].

3.2.5 Hot interstellar medium (thermal bremsstrahlung, soft X-rays)

Ionization: negligible. Heating: significant, and possibly the dominant *soft-band* contributor. Chandra observations of nearby star-forming galaxies show ubiquitous sub-keV thermal ISM emission that generally dominates the soft X-ray luminosity — precisely the band that interacts with the IGM. Pacucci et al. [46] show that if the first galaxies’ soft band is ISM-dominated rather than XRB-dominated, IGM temperature fluctuations are substantially increased and the large-scale 21-cm power is higher by roughly a factor of three. Eide et al. [27] independently find that the ISM component (and to a lesser extent XRBs) significantly shapes the thermal and ionization state of the neutral and lowly-ionized hydrogen, and thereby controls the 21-cm absorption-to-emission transition.

3.2.6 Cosmic rays

Ionization: minor, and poorly constrained. Tueros et al. [53] simulate particle cascades in the primordial IGM and argue that cosmic rays accelerated at the termination shocks of the first microquasar jets could contribute to reionization *as much as* the X-rays from the same objects’ accretion discs, depending on the electron-to-proton ratio; if the primordial magnetic field is

$\sim 10^{-17}$ G the cosmic rays can ionize material far from the first stars. [This is a single-paper, model-dependent claim; it is not corroborated elsewhere in the corpus.]

Heating: potentially significant at cosmic dawn. Sazonov and Sunyaev [54] argue that Pop III supernova remnants can channel a significant fraction of their kinetic energy into low-energy ($E < 30$ MeV) cosmic rays that escape into the IGM and heat it by ~ 10 – 100 K by $z \sim 15$, *before* X-ray mechanisms take over. Gessey-Jones et al. [47] build the first 3D semi-numerical model of this channel and find that cosmic-ray heating is short-ranged, hence clustered around star-forming sites, producing a sharper contrast between heated (emission) and unheated (absorption) regions and therefore *more small-scale 21-cm power* than X-ray heating — an observational discriminant. In models where cosmic rays escape efficiently only from low-mass haloes (as expected for Pop III remnants), the interplay with Lyman–Werner feedback can produce a local peak in T_K and a flattened global absorption trough.

3.2.7 Ly α photons: Wouthuysen–Field coupling, Ly α and CMB heating

Ionization: negligible (Ly α photons are below the ionization threshold). **Heating: subdominant in general, but decisive for the interpretation of the 21-cm signal.** Reis et al. [55] include Ly α heating, CMB heating mediated by Ly α photons, and multiple Ly α scattering, and find:

- In models with weak X-ray heating these processes raise the gas temperature by up to $\mathcal{O}(100)$ K.
- The global-signal absorption trough acquires a *floor* of -165 mK at $z \approx 15$ – 19 , whereas for a purely adiabatically cooling universe the deepest possible absorption is -178 mK at $z = 19$, -216 mK at $z = 15$ and -264 mK at $z = 10$.
- The maximum possible power at $z = 9$, $k = 0.1$ Mpc $^{-1}$ is attenuated by a factor 6.6.
- At cosmic dawn the heating terms are subdominant, but multiple scattering amplifies the power spectrum by a factor ~ 2 – 5 .

Crucially, Ly α photons are what couple T_S to T_K in the first place, so *no* 21-cm constraint on any other heating agent exists without them.

3.2.8 Secondary electrons (a channel, not a source)

Not an agent in its own right, but the mechanism through which every high-energy agent (X-rays, cosmic rays, and any exotic injection) is converted into ionization, heat and Ly α photons. Furlanetto and Stoeve [45] provide the standard energy-deposition fractions; their key results are summarised in Section 3.1 (Criterion 3). Any statement about the relative ionizing versus heating power of a hard-photon source is meaningless without specifying $f_{\text{ion}}(E, x_e)$ and $f_{\text{heat}}(E, x_e)$.

3.2.9 Structure-formation shocks

Ionization: negligible. Heating: guaranteed, and dominant before the first sources switch on. Gravitational collapse drives shocks into the IGM irrespective of any radiative source, so this channel has no free efficiency parameter — it is the floor beneath every other heating mechanism. Furlanetto and Loeb [56] extend the Press–Schechter formalism to large-scale structure shocks and show that *before* reionization the thermal energy of the IGM is dominated by these shocks rather than by virialised objects; the same shocks strip $\sim 10\%$ of the gas from dark-matter minihaloes, accelerate cosmic rays (linking to Section 3), and generate a diffuse inverse-Compton/cooling radiation background *before the first stars form*.

Quantitatively the effect on the observable is modest but not ignorable. Running high-resolution hydrodynamic simulations of early structure formation with star formation and feedback switched off, Xu et al. [57] find that the combination of non-linear density fluctuations, shock heating and Compton heating reduces the *maximum* global 21-cm absorption depth by $\sim 15\%$ at $z =$

17 relative to a homogeneous or linearly-fluctuating IGM. This matters directly for claims of anomalously deep absorption: a 15% bias must be removed before invoking new physics.

3.2.10 An excess radio background (a background, not a source of energy)

Ionization: none. Heating: none — but it changes every 21-cm inference. This entry is included because it is the principal *degeneracy* affecting the constraints of Sections 2.6 and 2.7: since $\delta T_b \propto (1 - T_{\text{rad}}/T_S)$, an excess radio background at 1.42 GHz mimics a colder IGM, so a 21-cm upper limit that appears to require heating can instead be satisfied by extra radio photons.

- **Phenomenology.** Assuming a uniform synchrotron-like excess, Fialkov and Barkana [58] show that it deepens the global signal and *enhances* the power spectrum; matching the reported EDGES Low-Band feature together with EDGES High-Band, high- z quasar and extragalactic radio-background limits requires a star-formation efficiency $\geq 2.8\%$, a minimum star-forming halo mass $M_h \lesssim 10^9 M_\odot$ at $z = 17$, and an excess of at least $1.9\times$ the CMB at 78 MHz (i.e. 0.1% of the CMB at 1.42 GHz). Notably, the X-ray source properties remain *unconstrained* in such models — which is precisely why this degeneracy weakens every heating-agent inference.
- **A candidate physical source.** Reis et al. [59] drop the uniformity assumption and model an inhomogeneous background from high- z radio galaxies. This introduces a genuinely new type of 21-cm fluctuation, alongside density, velocity, Ly α coupling, heating and reionization; even a moderately enhanced radio efficiency significantly alters the power spectrum in weak-X-ray-heating models, and EDGES-compatible models enhance the $z \sim 17$ power spectrum by up to two orders of magnitude.
- **A strong counter-argument.** Sharma [60] points out that for the relativistic electrons invoked in these scenarios (accelerated by accretion onto SMBHs or by first-star supernovae), the synchrotron or inverse-Compton cooling time is at least three orders of magnitude shorter than the duration of the EDGES feature, irrespective of magnetic field strength; the resulting 1.42 GHz emission is $\sim 10^3$ times smaller than the non-cooling estimate. On this argument astrophysical excess-radio-background scenarios are “very unlikely”.

Status: unresolved. The corpus contains both a viable phenomenology [58, 59] and a strong energetics objection [60], and the LOFAR analysis of Ghara et al. [28] leaves the radio-background efficiency unconstrained over its full prior at 95% credibility. Every T_S or T_K limit quoted in this document should be read as conditional on there being no such background.

3.3 Summary table of agents

Table 5: Contribution of each agent to hydrogen ionization and to IGM heating. Thresholds follow the convention of Section 3.1 (dominant $> 50\%$, significant 10–50%, minor 1–10%, negligible $< 1\%$); words in quotes are the source paper’s own.

Agent	H ionization	IGM heating	Key references
Star-forming galaxies (LyC)	Dominant	Dominant for ionized phase ($\sim 1\text{--}2 \times 10^4$ K); negligible for neutral IGM	Naidu et al. [6], Robertson et al. [8], Giovinazzo et al. [22], Finkelstein et al. [23], Eide et al. [27], Atek et al. [43]
– bright “oligarchs”	$> 80\%$ of budget in one model; contested	as above	Naidu et al. [6], Giovinazzo et al. [22]

Agent	H ionization	IGM heating	Key references
– faint dwarfs	Dominant in other models; contested	as above	Finkelstein et al. [23], Atek et al. [43]
Population III stars	Negligible at $z \lesssim 10$ ($< 1\%$ at $z = 10$; $\lesssim 10\%$ bound from DLA abundances); $\gtrsim 50\%$ only at $z \gtrsim 30$	Indirect, via Pop III SN cosmic rays and Pop III XRBs	Murphy et al. [33], Dasgupta et al. [48], Kulkarni et al. [49], Sazonov and Sunyaev [54]
Quasars / luminous AGN	Minor: $< 3\%$ of $\langle \Gamma_{\text{HI}} \rangle$ at $z \sim 6$ ($\sim 10\%$ if LF extended to $M_{1450} = -18$)	Dominant <i>after</i> H reionization, via He II	Kulkarni et al. [38], Madau and Haardt [50], Dayal et al. [51]
Faint AGN / little red dots	Contested: 17% works, AGN-only fails [39]; 31–75% claimed [52]	Overheats IGM at $z \lesssim 5$ if dominant	Asthana et al. [39], Dayal et al. [51], Singha et al. [52]
AGN, He II ionization	Dominant (only viable source of ≥ 54.4 eV photons)	Dominant: $+ \sim 10^4$ K locally; sets T_0 peak at $z \sim 3$	Villasenor et al. [26], Eide et al. [27], Kulkarni et al. [38], Madau and Haardt [50]
High-mass X-ray binaries	Negligible (x_e never exceeds 1%)	Leading candidate; requires low-metallicity L_X/SFR	HERA Collaboration et al. [1], Madau and Fragos [37], Dasgupta et al. [48]
Hot ISM (soft X-ray bremsstrahlung)	Negligible	Significant: can dominate the soft band; $\times 3$ large-scale 21-cm power	Eide et al. [27], Pacucci et al. [46]
Nuclear BHs / mini-quasars	Negligible (rare)	Significant locally; only sources fully ionizing He	Eide et al. [27]
Cosmic rays	Minor / unconstrained; comparable to microquasar X-rays in one model	Possibly significant at $z \sim 15$ ($\Delta T \sim 10\text{--}100$ K); distinctive small-scale 21-cm signature	Gessey-Jones et al. [47], Tueros et al. [53], Sazonov and Sunyaev [54]
$\text{Ly}\alpha$ photons (WF, $\text{Ly}\alpha/\text{CMB}$ heating)	Negligible (sub-threshold)	Up to $\mathcal{O}(100)$ K in weak-X-ray models; sets the -165 mK absorption floor; <i>required</i> for any 21-cm coupling	Reis et al. [55]
Structure-formation shocks	Negligible	Dominant before the first sources; reduces max. global absorption by $\sim 15\%$ at $z = 17$	Furlanetto and Loeb [56], Xu et al. [57]

Agent	H ionization	IGM heating	Key references
Secondary electrons	Channel, not source: $f_{\text{ion}}(E, x_e)$	Channel, not source: $f_{\text{heat}}(E, x_e)$; equal three-way split at 1–10 keV	Furlanetto and Stoeve [45]
Excess radio background	None	None — but mimics a cold IGM and so degrades every 21-cm heating constraint; energetics disputed	Ghara et al. [28], Fialkov and Barkana [58], Reis et al. [59], Sharma [60]
Small-scale structure / clumping	Not a source: a <i>sink</i> . $C \approx 12$ observationally vs $C \approx 3$ in simulations; $C_{\text{HII,rec}} = 6.2^{+4.1}_{-2.1}$ required	Sets the recombination rate and λ_{mfp} fluctuations; peak $C = 5$ –20 for ~ 10 Myr post-I-front	Davies et al. [40], Austin et al. [41], D’Aloisio et al. [44]

4 The parameters that govern the reionization history

This section collects the quantities that actually enter a reionization calculation — the ones you have to choose a value for before Eq. 3 can be integrated. Each parameter gets its own entry: the symbol and the range it spans across the corpus in the heading, then the individual determinations behind that range, one per bullet, each with its source.

Every bullet carries a tag: [M] for a value *measured* or inferred from data, [A] for one *assumed* as a model input, [D] for one *derived* here. The distinction matters — several of the widest ranges below are ranges of assumption, not of measurement. A heading range is the spread *across* the cited papers, never a confidence interval; where a single paper’s own uncertainty is quoted it appears in the bullet. Every value was checked against its source PDF under the audit of Section 1.3.

4.1 Source term — how many ionizing photons escape

f_{esc} — Lyman-continuum escape fraction of galaxies.	<i>corpus range:</i> 0.03–0.25
• [A] $< 5\%$ averaged through the bulk of the EoR, in a model where the faintest galaxies dominate the emissivity and ξ_{ion} rises towards fainter magnitudes [23]. • [A] $0.21^{+0.06}_{-0.04}$ for $M_{\text{UV}} < -13.5$, in the bright-galaxy picture, with $f_{\text{esc}} \propto \Sigma_{\text{SFR}}^{0.4 \pm 0.1}$ [6]. • [M] $\simeq 5\%$ inferred from the UV continuum slope β_{UV} via the low-redshift LzLCS calibration [41]. • [M] ≈ 7–10% in radiation-hydrodynamic simulations that include binary stars — about $3\times$ the single-star value [3]. • [A] 0.03–0.20 spanned as an explicit prior when apportioning the budget between galaxies and AGN [52]. $\hookrightarrow$ <i>Not observable at $z > 6$:</i> the IGM is opaque to exactly the photons one would need to detect [22]. Every value above is an inference.	

ξ_{ion} — Ionizing photon production efficiency, $\log_{10}(\text{Hz erg}^{-1})$.	<i>corpus range:</i> 25.05–25.80
• [A] 25.2, the canonical pre-JWST value [8, 42]. • [M] 25.80 ± 0.14 for eight ultra-faint lensed galaxies at $M_{\text{UV}} \sim -17$ to -15 — a factor ~ 4 above canonical [43]. • [M] 25.5–26.0 across the JWST-era compilation [42].	

- [M] $-0.006^{+0.019}_{-0.017} M_{\text{UV}} + 25.05^{+0.39}_{-0.34}$, i.e. *magnitude-independent* and consistent with the canonical HST value, from 1721 galaxies [41].
- [A] 25.7 adopted as an intermediate value [52].

ξ_{ion} — per unit SFR, $\log_{10}[\text{s}^{-1}/(\text{M}_{\odot} \text{yr}^{-1})]$. *corpus range:* 53.14

- [A] 53.14, fiducial, paired with $f_{\text{esc}} = 0.2$ [8].

α_{UV} — UV luminosity function faint-end slope. *corpus range:* -2.2 to -1.7

- [M] -2.2 ± 0.2 for star-forming galaxies and -1.7 ± 0.2 for “smouldering” ones — two populations with genuinely different slopes [41].

$M_{\text{UV}}^{\text{lim}}$ — Faint-end integration limit. *corpus range:* -17 to -11

- [M] -17 , the limit of direct observation [41].
- [A] -13.5 , the common working choice, set by the UVLF turnover in Abell S1063 [6, 41].
- [A] -11 , an extrapolation which only worsens the photon budget [42].

$\dot{n}_{\text{ion}}$ — Escaping ionizing emissivity ($\text{s}^{-1} \text{cMpc}^{-3}$). *corpus range:* $\simeq (0.3\text{--}0.7) \times 10^{51}$

- [M] $\simeq 0.6 \times 10^{51}$, remarkably flat across $4.9 \leq z \leq 6.0$, from forward-modelling the Ly α forest [7].
 - [M] $\log_{10} = 50.31^{+0.07}_{-0.06}$ for $M_{\text{UV}} < -17$ from JWST/NIRCam [41].
- $\hookrightarrow$ [D] Equivalent to 3.4 ionizing photons per hydrogen atom per Gyr (Section 1.4), against ≈ 3 photons per baryon required at $C \approx 12$ [40].

$f_{\star}$ — Star-formation efficiency. *corpus range:* $\geq 2.8\%$

- [M] $\geq 2.8\%$, a lower limit from matching the EDGES-era global signal together with high- z quasar and radio-background limits [58].

M_{min} — Minimum star-forming halo mass. *corpus range:* $10^9\text{--}10^{9.4} \text{M}_{\odot}$

- [M] $\gtrsim 2.6 \times 10^9 \text{M}_{\odot} [(1+z)/10]^{1/2}$, equivalently $V_c \gtrsim 50 \text{km s}^{-1}$ (95% credibility), from the joint 21-cm + Lyman-line + CMB fit [21].
- [A] $\log M_{\text{h}} \sim 9$ as the photosuppression mass during and after reionization [23].
- [M] $\lesssim 10^9 \text{M}_{\odot}$ at $z = 17$, an upper limit from the excess-radio-background fit [58].

4.2 Source term — the AGN contribution

f_{AGN} — AGN share of the hydrogen-ionizing budget. *corpus range:* $< 3\%$ to 75%

- [M] $< 3\%$ of $\langle \Gamma_{\text{HI}} \rangle$ at $z \sim 6$, rising to $\sim 10\%$ only if the luminosity function holds to $M_{1450} = -18$, from a homogenised sample of $> 80,000$ AGN [38].
- [M] 10–25% of the cumulative emissivity by $z = 4$ in models that match the reionization constraints [51].
- [M] 17% works: a QSO-assisted model fits the observed Ly α optical-depth distribution, while a QSO-only model does not [39].
- [M] 31–75% at $z > 5$ claimed from a unified rest-UV AGN luminosity function [52] — *disputed*; see the emissivity flag in Section 3.

β_{AGN} — AGN luminosity function faint-end slope. *corpus range:* -2.4 to -1.7

- [M] Steepens from -1.7 at $z < 2.2$ to -2.4 at $z \simeq 6$ [38].

α_{AGN} — AGN ionizing spectral index, $\epsilon_{\nu} \propto \nu^{-\alpha}$. *corpus range:* 1.7

- [A] 1.7, adopted as the AGN power law in radiative-transfer post-processing [39]. That paper prints it as $\alpha = -1.7$ in the $\nu^{+\alpha}$ convention; this document uses $\epsilon_\nu \propto \nu^{-\alpha}$ throughout, so a larger α is a *softer* spectrum.

t_Q — Quasar lifetime. *corpus range:* 10^3 – 10^8 yr

- [M] $\lesssim 7$ Myr across all redshift bins, fitted jointly with $\langle x_{\text{HI}} \rangle$ from stacked damping wings [16].
- [A] $10^3 \leq t_Q \leq 10^8$ yr marginalised over when extracting $\langle x_{\text{HI}} \rangle$ from individual quasars [18].
- [A] 10 Myr assumed for every AGN in the simulation [39].

4.3 Sink term — where the photons go

C_{HII} — Clumping factor, $\langle n_{\text{H}}^2 \rangle / \langle n_{\text{H}} \rangle^2$. *corpus range:* 3 to ≈ 12

- [A] 3, the long-standing fiducial [8].
- [M] Peaks at 5–20 about 10 Myr after the ionization front passes and relaxes to ≈ 3 only by 300 Myr; un-relaxed gas boosts total recombinations by $\approx 50\%$ [44].
- [M] $6.2_{-2.1}^{+4.1}$ *required* for stable reionization at $z \approx 6$ when the UVLF is extrapolated to $M_{\text{UV}} = -13.5$ [41].
- [M] ≈ 12 globally, from imposing ionization equilibrium on the measured λ_{mfp} and $\langle \Gamma_{\text{HI}} \rangle$ — four times the simulation value, and worth ≈ 3 photons per baryon [40].

λ_{mfp} — Mean free path of Lyman-continuum photons. *corpus range:* 0.75–9.09 pMpc

- [M] $9.09_{-1.28}^{+1.62}$ pMpc at $z = 5.1$ (68% confidence), on the low-redshift power-law extrapolation [24].
- [M] $0.75_{-0.45}^{+0.65}$ pMpc at $z = 6.0$ — a factor ~ 7 *below* that extrapolation; neglecting the quasar proximity effect biases it high by a factor ≥ 2 [24].
- [M] 8.3 – $57 h^{-1}$ cMpc over $6.0 \geq z \geq 4.9$, marginalised over $\langle \Gamma_{\text{HI}} \rangle$ and the thermal parameters [7].

α_B — Case-B recombination coefficient. *corpus range:* 1.43 – $2.59 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$

- [A] $2.59 \times 10^{-13} \text{ cm}^3 \text{ s}^{-1}$ at $T = 10^4 \text{ K}$ and 1.43×10^{-13} at $T = 2 \times 10^4 \text{ K}$ — the temperature Robertson et al. [8] adopt. Tabulated values from Osterbrock & Ferland (2006), Table 2.1, a textbook and not part of the corpus.
- ↪ Case B excludes recombinations directly to the ground state, which emit another ionizing photon and so cause no net change; the choice alone is worth $\sim 40\%$ against case A.

t_{rec} — IGM recombination time at $z = 7$. *corpus range:* 704–2113 Myr

- [D] 704 Myr for $C_{\text{HII}} = 3$ and 2113 Myr for $C_{\text{HII}} = 1$, at $T = 2 \times 10^4 \text{ K}$ (Section 1.4). The factor 3 in C_{HII} maps directly onto the recombination sink.

4.4 The resulting history

τ — CMB Thomson optical depth. *corpus range:* 0.054–0.067

- [M] 0.054 ± 0.007 (68% CL), TT,TE,EE+lowE+lensing [5].
- [M] $0.0566_{-0.0062}^{+0.0053}$ from large-scale polarization alone and 0.059 ± 0.006 combined, with improved HFI map-making [9].
- [M] 0.067 ± 0.011 from an astrophysics-anchored fit that excludes large-scale CMB polarization [10].

z_{50} — Mid-point, $\langle x_{\text{HI}} \rangle = 0.5$. *corpus range:* 6.9–8.1

- [M] $7.16^{+0.15}_{-0.12}$ from the joint 21-cm + Lyman-line + CMB analysis [21].
- [M] $6.9 < z < 7.6$ from quasar absorption [20].
- [A] 8.14 ± 0.61 — but this is a *tanh-model* quantity, not a direct measurement [9].

z_{end} — Completion redshift. *corpus range:* 5.2–5.3

- [M] Homogeneous-UVB models ruled out at $z \geq 5.4$ ($> 3.5\sigma$) from 67 quasar sightlines [14].
- [M] Long dark gaps persist to $z \simeq 5.3$ [15].
- [M] Radiative-transfer modelling of $\langle \Gamma_{\text{HI}} \rangle$, λ_{mfp} and $\langle x_{\text{HI}} \rangle$ favours completion by $z \simeq 5.2$ [7].

Δz_{re} — Duration. *corpus range:* < 1.8 to < 4.2

- [M] < 1.8 (95% credibility) from 21-cm + Lyman-line + CMB [21].
- [M] $1.0^{+1.6}_{-0.7}$, and < 4.1 at 95% confidence, from the first $\geq 3\sigma$ kSZ detection [31].
- [M] $\Delta^{50} z_{\text{re}} < 4.2$ (95%) from SPT-3G, measured between ionized fractions of 25% and 75% — a different interval from the row above, so the two limits are not directly comparable [2].

$\langle \Gamma_{\text{HI}} \rangle$ — H I photoionization rate. *corpus range:* $(0.145\text{--}0.557) \times 10^{-12} \text{ s}^{-1}$

- [M] Falls from $0.557^{+0.376}_{-0.218} \times 10^{-12} \text{ s}^{-1}$ at $z = 5.0$ to $0.145^{+0.157}_{-0.087} \times 10^{-12} \text{ s}^{-1}$ at $z = 6.0$ — only a factor ~ 4 , far less than the four orders of magnitude in $\langle x_{\text{HI}} \rangle$ over the same interval, because the collapse is driven by λ_{mfp} rather than by the sources [7].

4.5 Thermal state and heating agents

T_0 — Ionized IGM temperature at mean density. *corpus range:* $(1.05\text{--}1.4) \times 10^4 \text{ K}$

- [M] 11000 ± 1600 , 10500 ± 2100 and $12000 \pm 2200 \text{ K}$ at $z = 5.4$, 5.6 , 5.8 , from transmission-spike widths [25].
- [M] Two peaks: $1.3 \times 10^4 \text{ K}$ when hydrogen reionization completes near $z \sim 6$, and $1.4 \times 10^4 \text{ K}$ when He II completes near $z \sim 3$ [26].

$\hookrightarrow$ Measures the *ionized* phase only. Never plot it on the same curve as T_{S} .

T_{S} — Spin temperature of the neutral IGM. *corpus range:* $> 6\text{--}11 \text{ K}$

- [M] $> 11.0 \text{ K}$ at $z = 7.9$ and $> 6.2 \text{ K}$ at $z = 10.4$ (95% credibility; > 35.2 and $> 26.4 \text{ K}$ at 68%), ruling out cold reionization [1].

$\langle T_{\text{HI}} \rangle$ — Kinetic temperature of the neutral IGM. *corpus range:* $\gtrsim 27 \text{ K}$

- [M] Models with $\langle T_{\text{HI}} \rangle \lesssim 27 \text{ K}$ at $z \approx 5.6$ for $\langle x_{\text{HI}} \rangle = 0.1$ are disfavoured at 68% credibility, from a null detection of the 21-cm forest [30].

$L_{X<2\text{keV}}/\text{SFR}$ — X-ray heating efficiency, $\text{ergs}^{-1}(\text{M}_{\odot} \text{ yr}^{-1})^{-1}$. *corpus range:* $10^{39.9}\text{--}10^{41.8}$

- [M] $10^{39.9}\text{--}10^{41.6}$ (68% HPD), tightening to $10^{40.4}\text{--}10^{41.8}$ at 95%. More than 99% of the posterior excludes the local, metal-enriched relation, requiring low-metallicity high-mass X-ray binaries [1].

f_{heat} — Fraction of fast-electron energy deposited as heat. *corpus range:* $1/3$ to 1

- [D] $\rightarrow 1$ below 10.2 eV , where an electron can neither excite nor ionize hydrogen; $\approx 1/3$ at $1\text{--}10 \text{ keV}$, where heating, ionization and excitation split roughly equally; strongly dependent on x_{e} and almost independent of density [45].

ΔT_{CR} — Cosmic-ray heating. *corpus range:* $\sim 10\text{--}100 \text{ K}$

- [A] $\sim 10\text{--}100 \text{ K}$ by $z \sim 15$ from sub-relativistic ($E < 30 \text{ MeV}$) cosmic rays escaping Pop III supernova remnants [54].

$\Delta T_{\text{Ly}\alpha}$ — Ly α + CMB heating. *corpus range:* up to $\mathcal{O}(100)$ K

- [A] Up to $\mathcal{O}(100)$ K in models with weak X-ray heating, raising the global-signal absorption floor to -165 mK [55].

A_r — Excess radio background. *corpus range:* $\geq 1.9\times$ CMB at 78 MHz

- [A] $\geq 1.9\times$ the CMB at 78 MHz, equivalently 0.1% of the CMB at 1.42 GHz, if the EDGES feature is to be explained [58]. Energetically disputed [60] and observationally unconstrained over its full prior [28].

4.6 Cosmological inputs (held fixed)

— — Cosmology. *corpus range:* Planck 2018

- [M] $\Omega_b h^2 = 0.02237 \pm 0.00015$; $h = 0.6736$ (from $H_0 = 67.36 \pm 0.54 \text{ km s}^{-1} \text{ Mpc}^{-1}$); $Y_p = 0.2454$ [5].
- [D] $\langle n_H \rangle = 1.895 \times 10^{-7} \text{ cm}^{-3}$ comoving, and $n_{\text{He}}/n_{\text{H}} = 0.0813$ (Section 1.4).

4.7 Reading these numbers: the degeneracies

The parameters above are not independent, and the degeneracies are what make reionization hard. Quoting any one of them without saying which values of the others were assumed alongside it is close to meaningless. Nine are stated explicitly in the corpus:

1. $f_{\text{esc}} \times \xi_{\text{ion}}$ — **the ionizing budget.** Only the product sets the timing: “increasing ξ_{ion} per galaxy while decreasing f_{esc} will yield identical effects on the IGM”, so the two are “at face value, fully degenerate” [42]. Direct Balmer-line measurements of ξ_{ion} break it partially; clustering of reionization bubbles, e.g. with 21-cm, is what would break it fully.
2. $A_s \leftrightarrow \tau$ — **the CMB’s own degeneracy.** High- ℓ CMB spectra measure the combination $A_s e^{-2\tau}$, so $\delta A_s / A_s = 2 \delta \tau$ [9]. Large-scale polarization is the *only* thing that breaks it and delivers τ directly — which is precisely why excluding low- ℓ EE data sends τ to ~ 0.09 and destabilises the cosmology [10]. Planck quote $A_s e^{-2\tau} = (1.884 \pm 0.012) \times 10^{-9}$ [5].
3. $C_{\text{HII}} \leftrightarrow$ **source term.** Clumping enters only through t_{rec} , so it trades directly against $\dot{n}_{\text{ion}}$. Going from $C = 3$ to $C \approx 12$ demands roughly twice as many ionizing photons [40], which is why the clumping dispute and the photon-budget crisis are the same argument seen from two ends [41].
4. $\langle \Gamma_{\text{HI}} \rangle \leftrightarrow \lambda_{\text{mfp}}$. The forest constrains the product, not either alone: “it is important to vary both $\lambda_{\text{mfp,HI}}$ and $\langle \Gamma_{\text{HI}} \rangle$ ” when matching the τ_{eff} distribution, and the quoted $\langle \Gamma_{\text{HI}} \rangle$ errors are “larger and more realistic” precisely because they are marginalised over λ_{mfp} [7]. There is a third axis: $\langle \Gamma_{\text{HI}} \rangle$ is also degenerate with the thermal parameters, which is why spike *widths* rather than depths are used to isolate T_0 [25].
5. $\langle x_{\text{HI}} \rangle \leftrightarrow t_Q$ (**damping wings**). Quantitative constraints are “challenging due to ... degeneracies with quasar lifetime” [18]; both papers using this probe therefore fit $\langle x_{\text{HI}} \rangle$ and t_Q *jointly* [16].
6. $\langle x_{\text{HI}} \rangle \leftrightarrow R_b$ (**bubble size**). In JWST galaxy damping wings the two are correlated because strong Ly α emission can mimic high transmission and hence a large ionized bubble, “which leads to the degeneracy between the two parameters and the large uncertainty” [19].

7. $A_r \leftrightarrow f_X$ — **radio background versus X-ray heating.** Both set the depth of the 21-cm absorption feature, so they are degenerate [58]; and f_X is in turn degenerate with $f_\star$, since X-ray heating and the Ly α background scale together [55]. This is why every T_S limit in Section 2.6 is conditional, and why the radio-background efficiency remains unconstrained over its whole prior [28]. Cosmic-ray heating joins the same tangle: its 21-cm signature “may be degenerate with those of other physical processes” [47].
8. **Dark matter mass $\leftrightarrow$ heating history.** In the 21-cm forest, small-scale power is suppressed both by warm dark matter and by early heating, and the two “are degenerate”; the 1D power spectrum is proposed specifically to break it [29].
9. **tSZ $\leftrightarrow$ kSZ $\leftrightarrow$ CIB.** The tSZ–CIB correlation is “partially degenerate with the tSZ and kSZ power” [31], with “clear degeneracies between all three parameters” [2]. Since Δz_{re} is inferred from the patchy kSZ amplitude after subtracting an *assumed* homogeneous component, the duration constraint inherits all of it.

Three further couplings are worth knowing because they sit inside the simulations rather than in the fits: f_{esc} is degenerate with the star formation history [22], with the SFR and the stellar IMF [27], and with the feedback efficiency, “a natural outcome of the regulation of f_{esc} by feedback” [3]. A quoted f_{esc} is therefore a statement about a whole galaxy-formation model, not a property of galaxies.

5 Tensions and caveats worth carrying into the paper

1. **The photon budget crisis** (Figure 10). Taking JWST ξ_{ion} values and abundances at face value, the *already observed* galaxies ($M_{\text{UV}} < -15$) not only can drive reionization but would end it too early, in tension with the CMB τ and the Ly α forest; extending to $M_{\text{UV}} \approx -11$ makes it worse [42]. Proposed resolutions in the corpus: a milder $\xi_{\text{ion}}(z, M_{\text{UV}})$ evolution with two populations separated by burstiness, which for $M_{\text{UV}} \in [-20, -16]$ and a reasonable f_{esc} reproduces the $z \lesssim 6$ Ly α forest without overshooting [4]; and a leaking-minority picture in which $\sim 20\%$ of galaxies produce $\sim 87\%$ of the escaping photons [22].
2. **τ and its cosmological knock-on.** Excluding large-scale CMB polarization weakens the dynamical-dark-energy preference but pushes τ_{reio} up to ~ 0.09 ; anchoring on astrophysical $\langle x_{\text{HI}} \rangle$ data instead recovers $\tau_{\text{reio}} = 0.067 \pm 0.011$ and restores a $\gtrsim 2\sigma$ dynamical-dark-energy preference [10]. Reionization is thus no longer only an astrophysics problem.
3. **f_{esc} is not measurable at $z > 6$** [22]; every quoted value is a model inference, and models disagree by factors of a few ($< 5\%$ in Finkelstein et al. [23] versus 0.21 in Naidu et al. [6]).
4. **Simulation systematics dominate damping-wing $\langle x_{\text{HI}} \rangle$.** ATON versus CROC shift $\langle x_{\text{HI}} \rangle$ at $\langle z \rangle = 6.46$ from 0.21 to 0.57 [16].
5. **Bubble-size selection bias.** Bright galaxies sit in merged, atypically large ionized bubbles, up to a few dex above the cosmic average at the same $\langle x_{\text{HI}} \rangle$ [19].
6. **21-cm limits are exclusions of extremes, not measurements.** The LOFAR-disfavoured models are “extreme ones ... mainly driven by rare and large ionized or heated regions”, and the excess-radio-background parameter remains unconstrained over its full prior at 95% [28].
7. **T_0 from the forest and T_S from 21-cm are not the same quantity** (ionized versus neutral phase); they should never be plotted on the same thermal-history curve without saying so.
8. **An unreconciled numerical discrepancy** between the AGN emissivity of Singha et al. [52] and the total measured emissivity of Gaikwad et al. [7] — see the flag in Section 3.

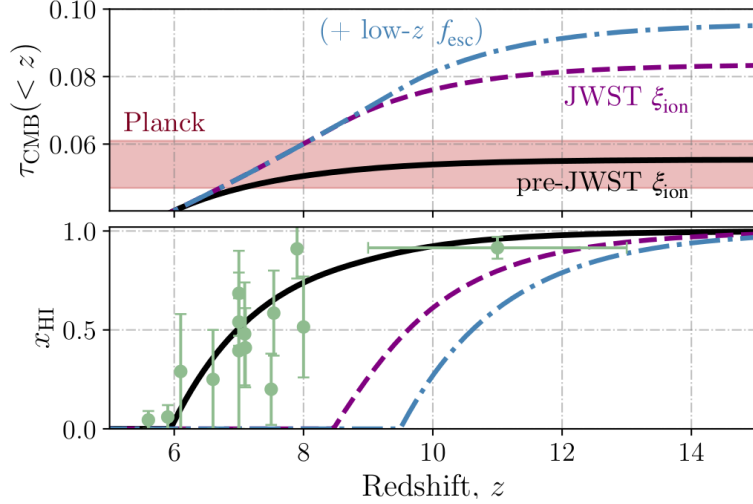


Figure 10: **The photon budget crisis.** Bottom: the neutral fraction $x_{\text{HI}}(z)$ for a pre-JWST model (black solid, $M_{\text{UV}}^{\text{cut}} = -13$, $f_{\text{esc}} = 0.2$), for the same model with a JWST-calibrated ξ_{ion} (purple dashed), and additionally with an f_{esc} from low- z analogues (blue dot-dashed); green points are the observational compilation. Top: the corresponding cumulative CMB optical depth against the *Planck* band (red). Taken at face value the JWST-calibrated models end reionization far too early and overshoot τ — the tension that Sections 3.1.2 and 5 discuss.

Reproduced from Muñoz et al. [42], Figure 1.

9. **The clumping factor is discrepant at the factor-of-4 level** between the observational estimate $C \approx 12$ [40] and the $C \approx 3$ relaxed value of radiation-hydrodynamic simulations [8, 44]. This is not a detail: it is worth a factor ≈ 2 in the required ionizing output, i.e. comparable in size and opposite in sign to the JWST ξ_{ion} excess. Austin et al. [41] argue the two cancel.
10. **ξ_{ion} itself is not settled.** Atek et al. [43] report $\log_{10} \xi_{\text{ion}} = 25.80 \pm 0.14$ for ultra-faint lensed galaxies, whereas Austin et al. [41], from a much larger stellar-mass-complete NIRCcam sample, find $\approx 25.05^{+0.39}_{-0.34}$ with *no* dependence on M_{UV} — consistent with the canonical HST value. The photon budget crisis is partly a disagreement about this one number.
11. **Can stellar populations make He II-ionizing photons?** The textbook answer is no, and the global budget does not need them to. But He II $\lambda 1640$ is now detected with high equivalent width at $z = 8.16$ and $z = 10.6$ [35, 36], the latter with no metal lines at all, and rotating Pop III models can reach the required hardness [34]. Meanwhile the simulations that report stars contributing nothing to He III adopt single-star spectral libraries [27]. Whether this is a genuine stellar contribution, an accreting black hole, or shocks is unresolved — the same ambiguity that attended CR7 — and it matters for *when* He II reionization starts, though not for who finishes it.
12. **Every 21-cm temperature limit is conditional on there being no excess radio background** (Section 3.2.10), a scenario that is phenomenologically viable [58, 59], energetically disputed [60], and observationally unconstrained [28].

6 Flagged for your decision (not included on my own judgement)

Per your instruction not to decide unilaterally, the following aspects and caveats surfaced during the search and are still *not* covered by the corpus. Each would require adding papers to the folders.

Resolved in this revision. Four topics previously listed here have now been incorporated at your request, with 12 papers added (**Agents**/ 25 $\rightarrow$ 32, **Constraints**/ 20 $\rightarrow$ 25): the excess radio background (Section 3.2.10); the clumping factor and small-scale structure (Section 3.1.2); structure-formation shock heating (Section 3); the 21-cm forest (Section 2.7); and kSZ patchy-reionization constraints (Section 2.8).

Still outstanding, awaiting your decision:

1. **Exotic energy injection as a heating/ionizing agent:** dark matter annihilation and decay, primordial black hole accretion and evaporation, and fuzzy/warm dark matter effects on the source population. You declined this in the first round; note that it now touches the document in two places anyway — Shao et al. [29] motivate the 21-cm forest power spectrum precisely as a way to break the warm-dark-matter/heating degeneracy, and Fialkov and Barkana [58] note that the X-ray source properties are unconstrained in excess-radio-background models. If you later want the agent catalogue to be exhaustive rather than astrophysical-only, this is the remaining gap.
2. **Gamma-ray bursts** as both probes of $\langle x_{\text{HI}} \rangle$ and (marginally) as sources.
3. **Metal absorption lines and the CGM** during the EoR as a complementary probe (mentioned in Fan et al. [20] but not developed here).

Per your instruction, citation counts are retained in Table 6 with the Semantic Scholar caveat of Section 1.2 stated in place.

A The corpus

All 60 PDFs were downloaded from arXiv on 8 September 2026 and verified (%PDF header, size > 20 kB). Filenames follow <FirstAuthorYear>_<slug>_arXiv<ID>.pdf. Three papers on He II $\lambda 1640$ were added on 10 September 2026 while checking the helium claim of Section 2.

Table 6: The 60-paper corpus. Folder A = **Agents**/, C = **Constraints**/.

Folder	First author	arXiv	Title
A	Furlanetto et al.	astro-ph/0312435	Small Scale Structure Shocks at Low and High Redshifts
A	Furlanetto et al.	0910.4410	Secondary ionization and heating by fast electrons
A	Kulkarni et al.	1310.0684	Chemical constraints on the contribution of Population III stars to cosmic reionization
A	Pacucci et al.	1403.6125	The X-ray spectra of the first galaxies: 21cm signatures
A	Tueros et al.	1409.6225	Cosmic reionization by primordial cosmic rays
A	Robertson et al.	1502.02024	Cosmic Reionization and Early Star-Forming Galaxies: A Joint Analysis of New Constraints from Planck and Hubble Space Telescope
A	Madau et al.	1507.07678	Cosmic Reionization after Planck: Could Quasars Do It All?
A	Sazonov et al.	1509.08408	Preheating of the Universe by cosmic rays from primordial supernovae at the beginning of cosmic reionization
A	Madau et al.	1606.07887	Radiation Backgrounds at Cosmic Dawn: X-Rays from Compact Binaries

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A	Rosdahl et al.	1801.07259	The SPHINX Cosmological Simulations of the First Billion Years: the Impact of Binary Stars on Reionization
A	Sharma	1804.05843	Astrophysical radio background cannot explain the EDGES 21-cm signal: constraints from cooling of non-thermal electrons
A	Kulkarni et al.	1807.09774	Evolution of the AGN UV luminosity function from redshift 7.5
A	Finkelstein et al.	1902.02792	Conditions for Reionizing the Universe with A Low Galaxy Ionizing Photon Escape Fraction
A	Fialkov et al.	1902.02438	Signature of Excess Radio Background in the 21-cm Global Signal and Power Spectrum
A	Naidu et al.	1907.13130	Rapid Reionization by the Oligarchs: The Case for Massive, UV-Bright, Star-Forming Galaxies with High Escape Fractions
A	Dayal et al.	2001.06021	Reionization with galaxies and active galactic nuclei
A	D'Aloisio et al.	2002.02467	Hydrodynamic Response of the Intergalactic Medium to Reionization
A	Reis et al.	2008.04315	High-redshift radio galaxies: a potential new source of 21-cm fluctuations
A	Eide et al.	2009.06631	Large scale simulations of H and He reionization and heating driven by stars and more energetic sources
A	Reis et al.	2101.01777	The subtlety of Ly-a photons: changing the expected range of the 21-cm signal
A	Xu et al.	2102.12865	Maximum Absorption of the Global 21 cm Spectrum in the Standard Cosmological Model
A	Murphy et al.	2105.06900	Ionizing photon production of Population III stars: effects of rotation, convection, and initial mass function
A	Kannan et al.	2110.00584	Introducing the THESAN project: radiation-magnetohydrodynamic simulations of the epoch of reionization
A	Wang et al.	2212.04476	A strong He II λ 1640 emitter with extremely blue UV spectral slope at $z=8.16$: presence of Pop III stars?
A	Gessey-Jones et al.	2304.07201	Signatures of Cosmic Ray Heating in 21-cm Observables
A	Atek et al.	2308.08540	Most of the photons that reionized the Universe came from dwarf galaxies
A	Munoz et al.	2404.07250	Reionization after JWST: a photon budget crisis?
A	Simmonds et al.	2409.01286	Ionising properties of galaxies in JADES for a stellar mass complete sample: resolving the cosmic ionising photon budget crisis at
A	Asthana et al.	2409.15453	The impact of faint AGN discovered by JWST on reionization
A	Wasserman et al.	2507.21764	Ultraviolet photon production rates of the first stars: Impact on the He II λ 1640 Å emission line from primordial star cl

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A	Singha et al.	2510.26990	Faint active galactic nuclei supplied 31-75% of hydrogen-ionizing photons at $z > 5$
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A	Maiolino et al.	2603.20362	The search for Population III: Confirmation of a HeII emitter with no metal lines at $z=10.6$
A	Dasgupta et al.	2604.11542	Impact of Stochastic Pop III X-ray Binaries on the Cosmological 21-cm Signal
A	Giovinazzo et al.	2607.22834	Reionization driven by the few: the ionizing budget of galaxies at $z=5-10$ from JWST/NIRSpec
C	Becker et al.	1407.4850	Evidence of patchy hydrogen reionization from an extreme Ly α trough below redshift six
C	McGreer et al.	1411.5375	Model-independent evidence in favor of an end to reionization by $z \approx 6$
C	Mason et al.	1709.05356	The Universe is Reionizing at $z \approx 7$: Bayesian Inference of the IGM Neutral Fraction Using Ly α Emission from Galaxies
C	Davies et al.	1802.06066	Quantitative Constraints on the Reionization History from the IGM Damping Wing Signature in Two Quasars at $z > 7$
C	HERA Collab. et al.	1807.06209	Planck 2018 results. VI. Cosmological parameters
C	Pagano et al.	1908.09856	Reionization optical depth determination from Planck HFI data with ten percent accuracy
C	Gaikwad et al.	2001.10018	Probing the thermal state of the intergalactic medium at $z > 5$ with the transmission spikes in high-resolution Ly α forest spectra
C	Reichardt et al.	2002.06197	An Improved Measurement of the Secondary Cosmic Microwave Background Anisotropies from the SPT-SZ + SPTpol Surveys
C	Becker et al.	2103.16610	The mean free path of ionizing photons at $5 < z < 6$: evidence for rapid evolution near reionization
C	Bosman et al.	2108.03699	Hydrogen reionisation ends by $z=5.3$: Lyman- α optical depth measured by the XQR-30 sample
C	Zhu et al.	2109.06295	Chasing the Tail of Cosmic Reionization with Dark Gap Statistics in the Ly α Forest over $5 < z < 6$
C	Villasenor et al.	2111.00019	Inferring the Thermal History of the Intergalactic Medium from the Properties of the Hydrogen and Helium Lyman-alpha Forest
C	HERA Collab. et al.	2210.04912	Improved Constraints on the 21 cm EoR Power Spectrum and the X-Ray Heating of the IGM with HERA Phase I Observations
C	Fan et al.	2212.06907	Quasars and the Intergalactic Medium at Cosmic Dawn
C	Gaikwad et al.	2304.02038	Measuring the photo-ionization rate, neutral fraction and mean free path of HI ionizing photons at $4.9 \leq z \leq 6.0$ from a large sample
C	Umeda et al.	2306.00487	JWST Measurements of Neutral Hydrogen Fractions and Ionized Bubble Sizes at $z=7-12$ Obtained with Ly α Damping Wing Absorption

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C	Shao et al.	2307.04130	The 21-cm forest as a simultaneous probe of dark matter and cosmic heating history
C	Durovcikova et al.	2401.10328	Chronicling the reionization history at 6 lesssim z lesssim 7 with emergent quasar damping wings
C	Davies et al.	2406.18186	The Predicament of Absorption-dominated Reionization II: Observational Estimate of the Clumping Factor at the End of Reionization
C	Sims et al.	2504.09725	Rapid and Late Cosmic Reionization Driven by Massive Galaxies: a Joint Analysis of Constraints from 21-cm, Lyman Line & CMB Data
C	Ghara et al.	2505.00373	Constraints on the state of the IGM at z sim 8-10 using redshifted 21-cm observations with LOFAR
C	Davies et al.	2510.25829	Updated dark pixel fraction constraints on reionization's end from the Lyman-series forests of XQR-30
C	Chaubal et al.	2601.20551	SPT-3G D1: A Measurement of Secondary Cosmic Microwave Background Anisotropy Power
C	Soltinsky et al.	2607.15341	Forest without Trees is still Fruitful: Constraints on the thermal state of the neutral IGM at z approx 5.6 with the 21-cm forest p
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